

# Sphenobasilar Synchondrosis (SBS) Strain Patterns

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Do.  
Make.  
Heal.  
Innovate.  
Reinvent the Future.

# Session Objectives

1. Student will be able to understand the principles and concepts of osteopathy the cranial field including the components of the primary respiratory mechanism (PRM).
2. Student will be able to understand the developmental anatomy and physiologic and non-physiologic motions that occur at the sphenobasilar synchondrosis (SBS).
3. Student will be able to understand the mechanisms and movement of the sphenoid and occiput that create cranial strain patterns and to be able to identify those patterns through a cranial vault hold.
4. Student will be able to understand cranial nerves function and identify potential presentations if there is nerve compression and the associated cranial bones that might be involved.
5. Student will understand the application and indications that osteopathy in the cranial field can be utilized for in different patient conditions and presentations.

# Case #1

- A 2-month-old baby girl is brought to the office with her mom complaining of her neck being twisted and a misshaped head.
- Mom reports that since delivery the baby does not like to turn her head to the right.
- She heard that osteopathic manipulative medicine can help her daughter and she has more questions for you before starting treatment.

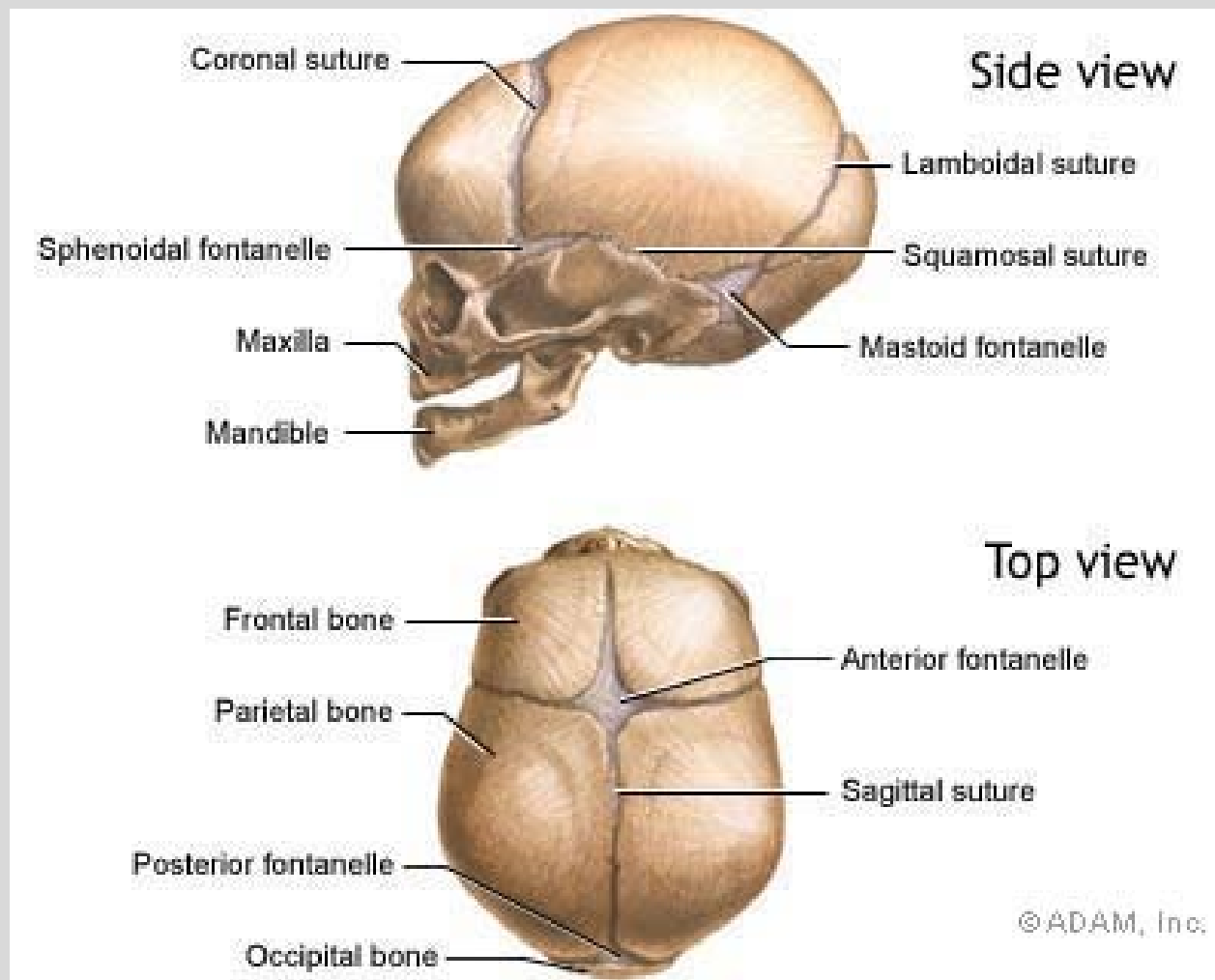
# Case #1

- How would you describe osteopathic manipulative medicine in the cranial field or Osteopathic Cranial Manipulative Medicine (OCMM)?
- How do you approach the patient?
- How do you explain to the mom through anatomy and physiology how osteopathy in the cranial field can help her baby?

# Application of Cranial Osteopathic Manipulative Medicine formally known as Osteopathy in the Cranial Field (OCF)

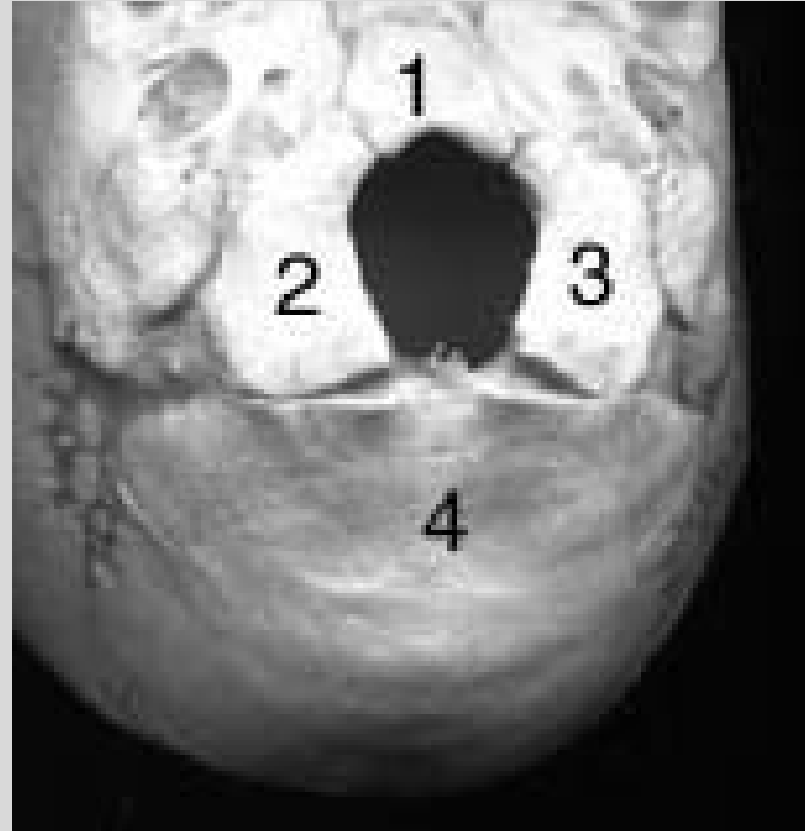
- History – Trauma
- Pediatrics – Delivery and Newborn considerations
- Observation
- Palpation
- Treatment

# Newborn skull

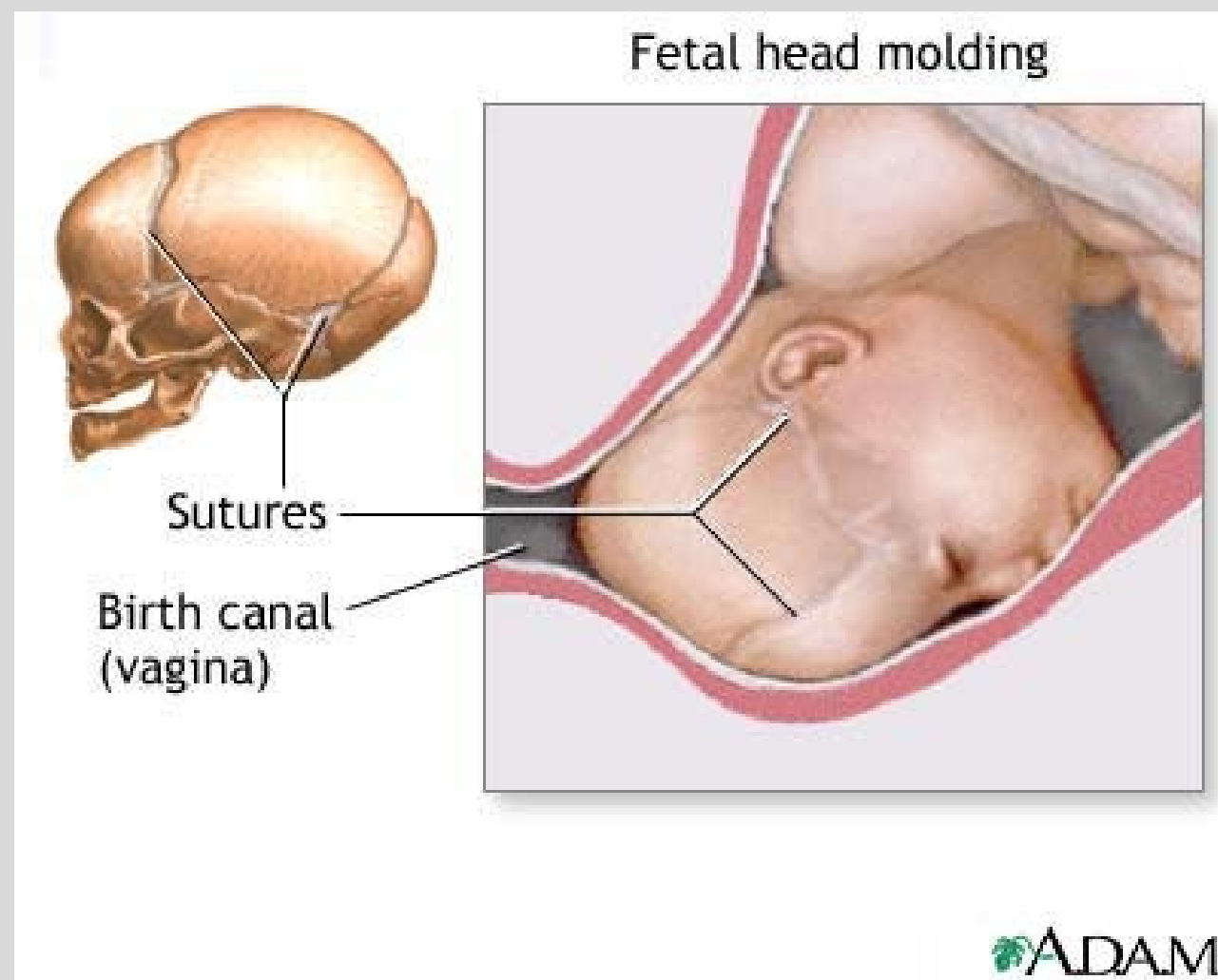


# Cranial Osteology - infant

- At birth:
- Occiput in 4 parts
- Sphenoid in 3 parts
- Temporal in 3 parts
- Maxillae in 2 parts
- 6 Fontanelles
- No mastoid process



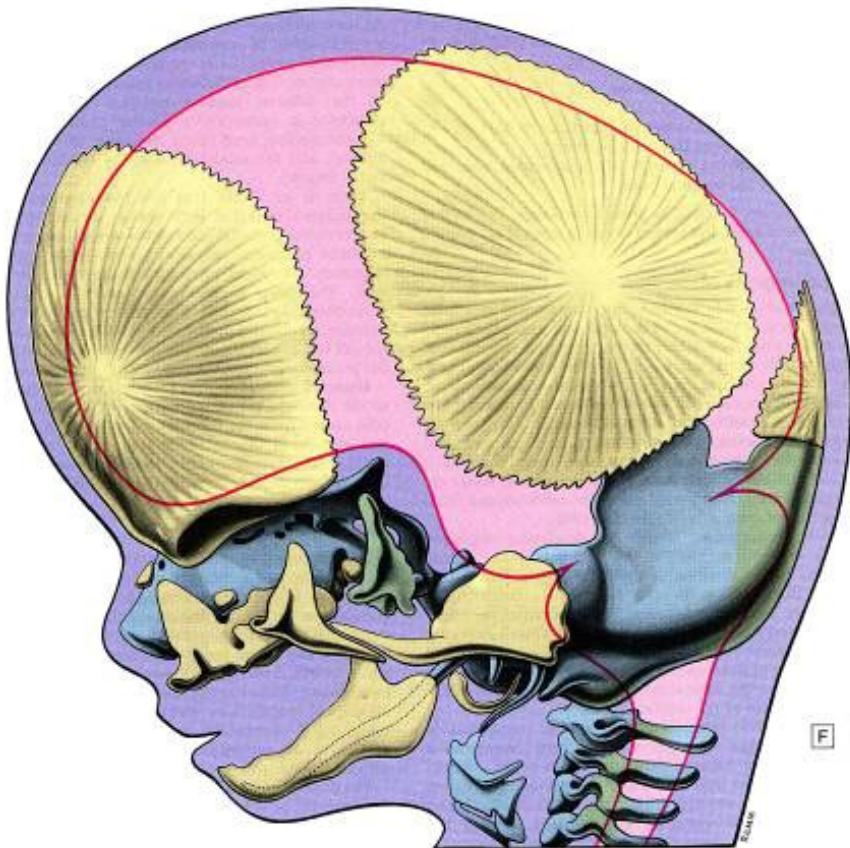
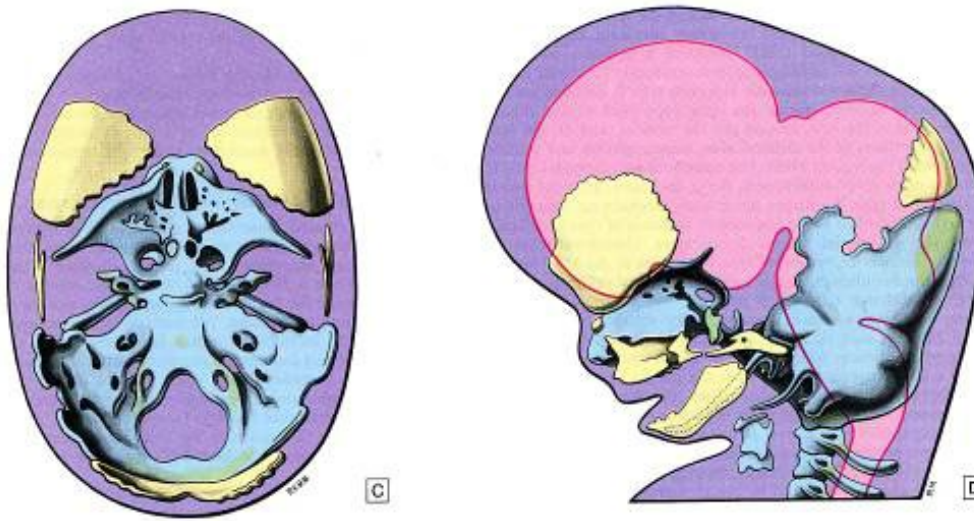
# Molding





# Cranial Bones

- The bones of the base are formed in cartilage.
- The bones of the vault are formed in membrane.





# Primary Respiratory Mechanism

1. Inherent *motility* of the brain and spinal cord.
2. Fluctuation of the CSF.
3. Movement of the intracranial and intraspinal membranes.
4. Articular *mobility* of the cranial bones.
5. Involuntary *mobility* of the sacrum between the ilia.

# Brain in Motion



<https://directorsblog.nih.gov/2018/06/21/brain-in-motion/>

# Cranial Contacts

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Vault



Fronto-occipital



Occipital (Becker)



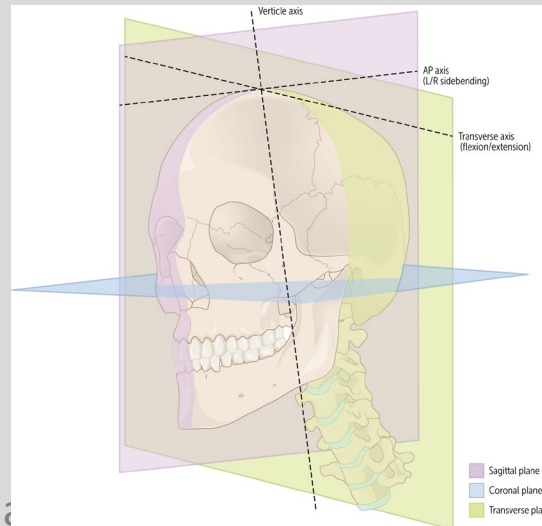
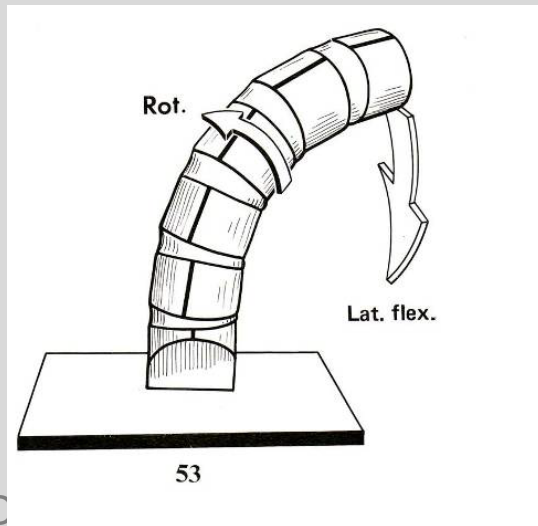
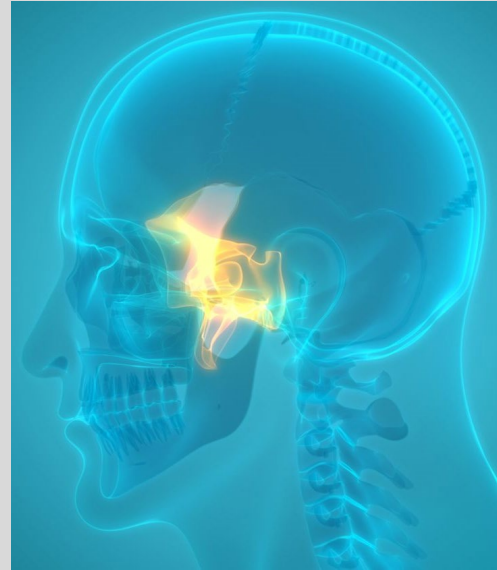
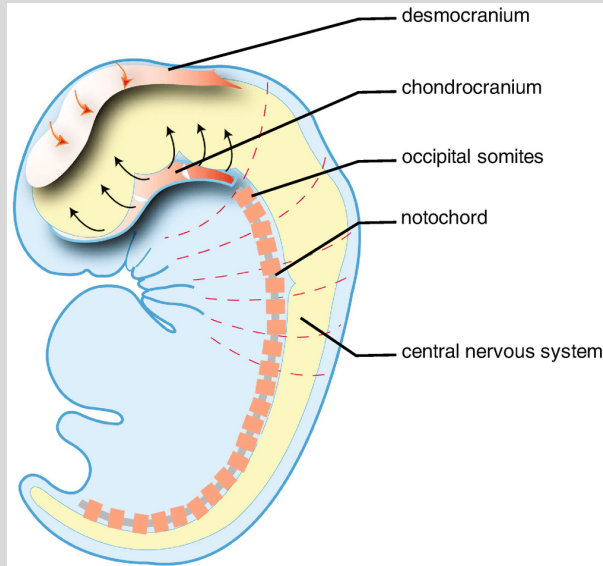


# Diagnosis



Sensing what is happening at the cranial base by palpation of the vault

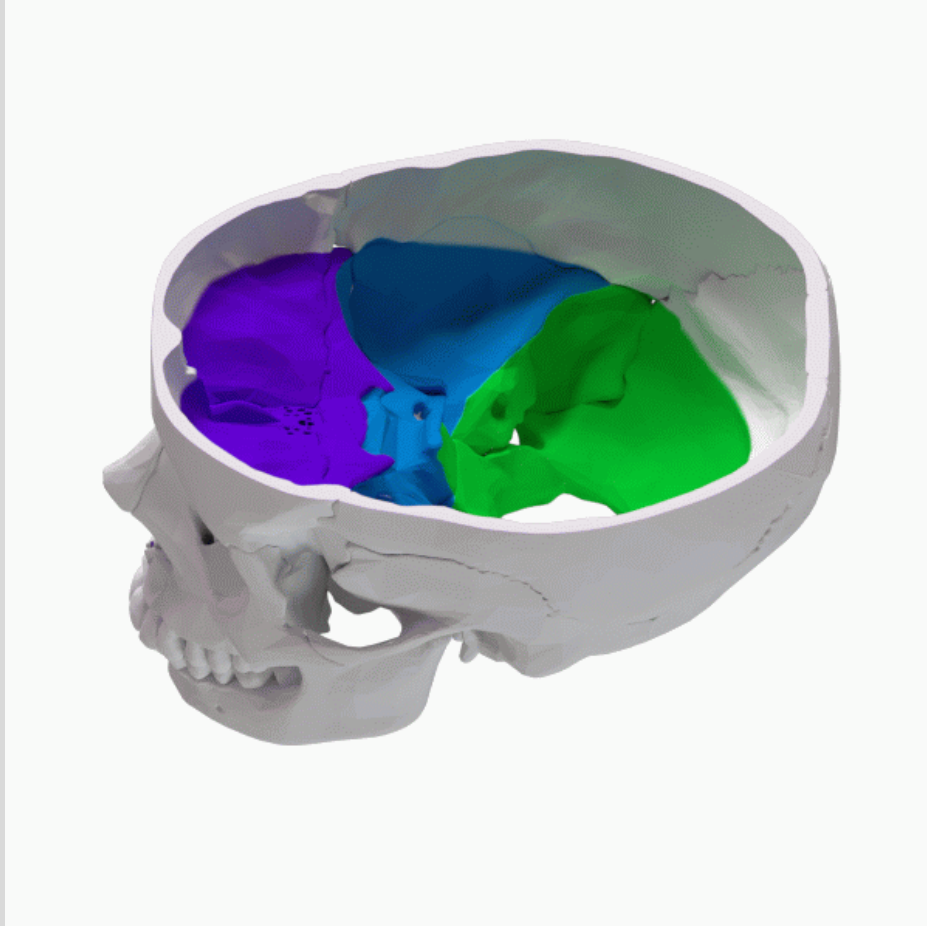
# Development of the SBS



- The SBS is the articulation between the sphenoid and occiput.
- Forms from cartilage dubbed "cranial vertebra"
- Continuation of the spine with a change in orientation
- Similar motion can occur at the SBS as in the spine (flexion/ extension, rotation and side bending)



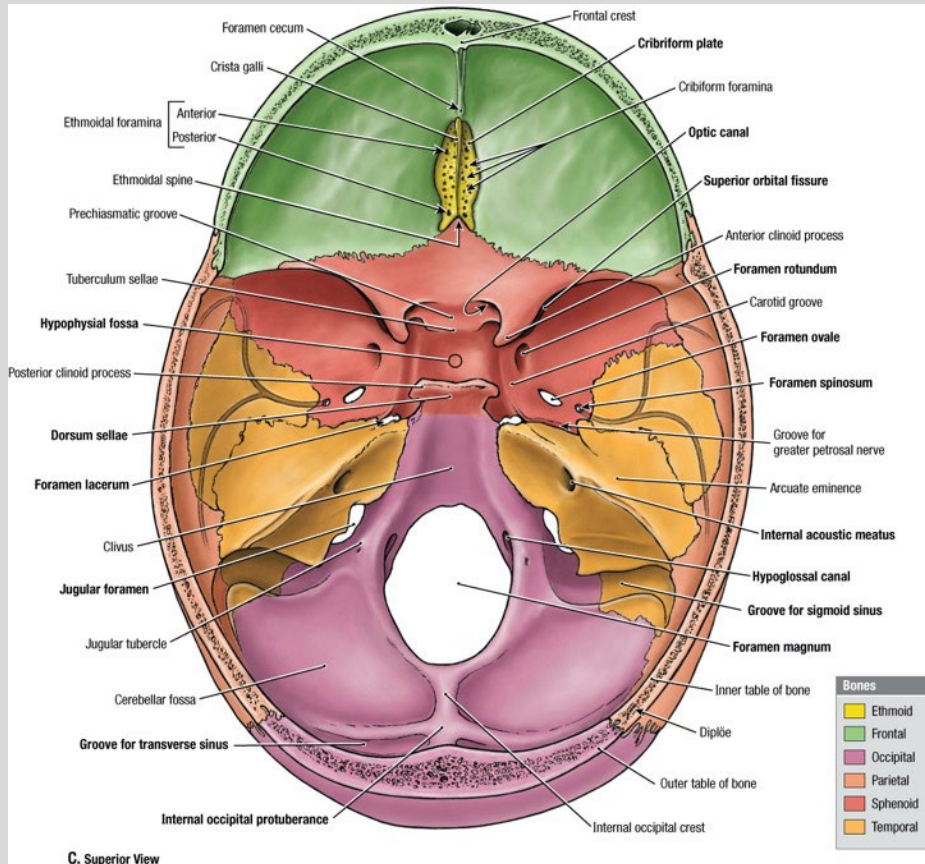
# The Base - Sphenobasilar Junction (SBS)



- All motion patterns are described in reference to the SBS.
- The SBS lies at the center of the cranial base, and therefore makes an excellent reference point.
- The cranium is designed so as to direct traumatic forces to the center, where the forces can be absorbed

[https://commons.wikimedia.org/wiki/File:Cranial\\_fossae\\_-\\_animation.gif](https://commons.wikimedia.org/wiki/File:Cranial_fossae_-_animation.gif)

# Strain patterns – etiology

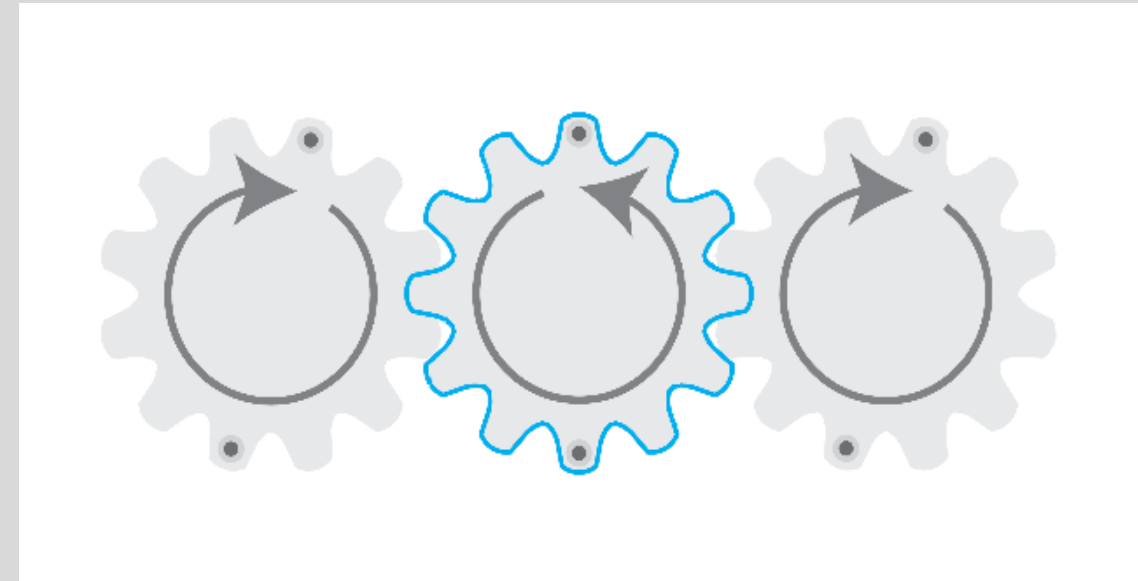
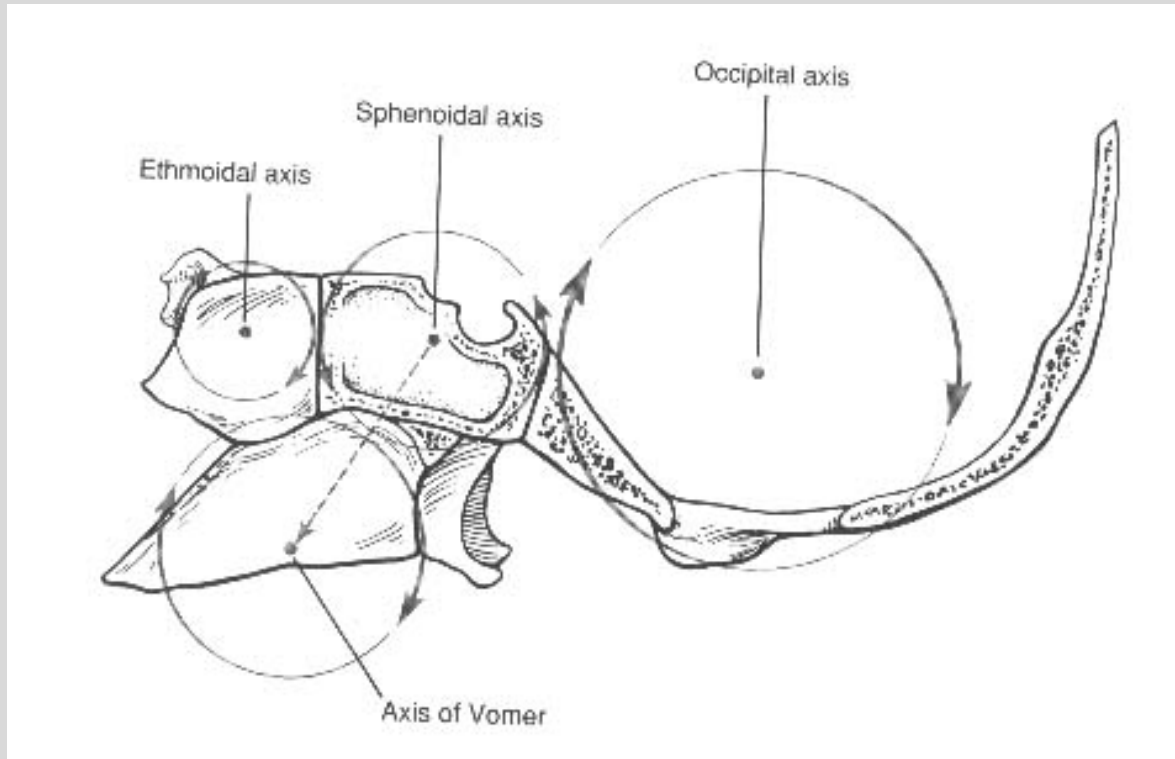


- Traumatic forces produce strain within the base, focused at the SBS and surrounding area, and this will produce an abnormal motion pattern.
- Sutherland through observations on himself and patients.
- Describes these cranial patterns as a schematic to allow for systematic evaluation and documentation
- Most patients will have a predominate pattern due to birth or subsequent patterns





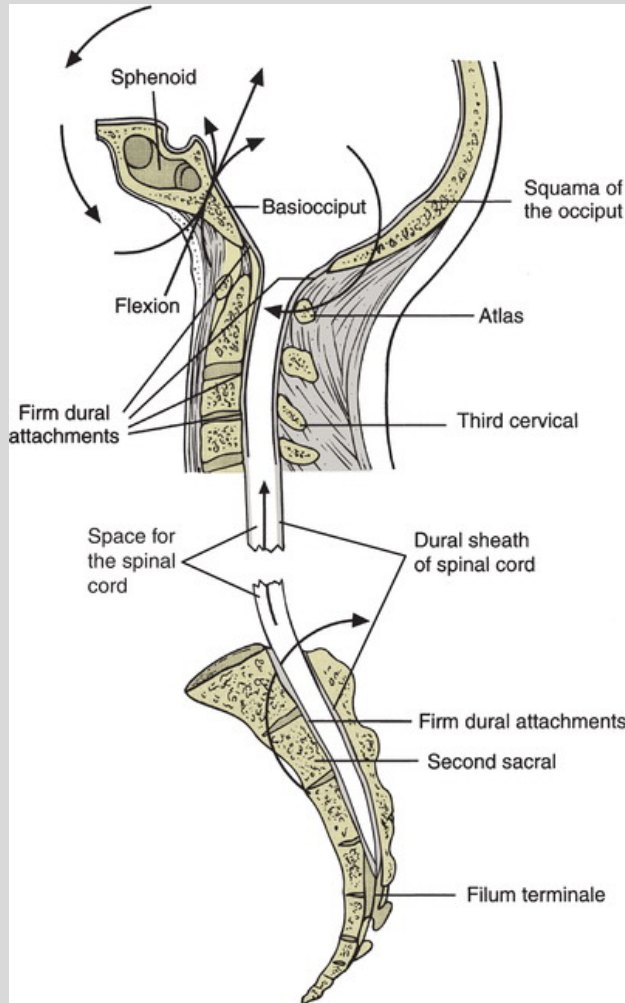
# Movement of the Cranial Base



- The midline bones move in a gearlike manner around transverse axes. This motion is called flexion (inhalation phase) and extension (exhalation phase)



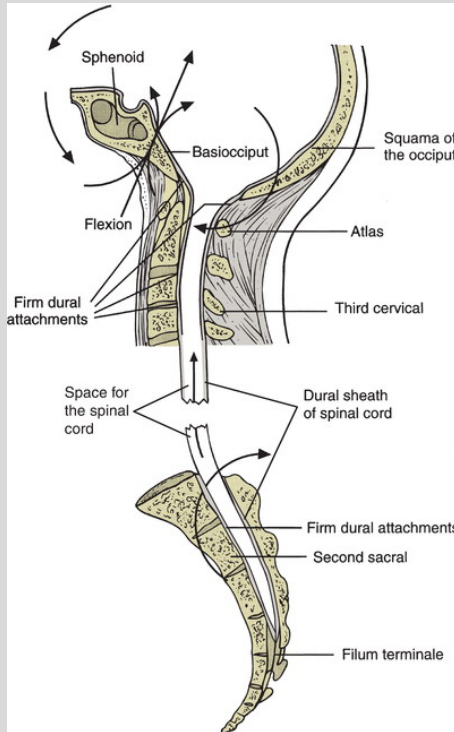
# Key dural connections - Core link



- The cranial dura continues down the spinal canal and attaches to C2&C3 and the sacrum
- With craniosacral flexion the sacrum counternutates (or rotates posteriorly)
- With craniosacral extension the sacrum nutates (rotates anteriorly)

From: **38 Osteopathic Cranial Manipulative Medicine**

Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e, 2018



**Legend:**

Craniosacral mechanism with arrows indicating direction of motion during flexion phase of physiologic motion.

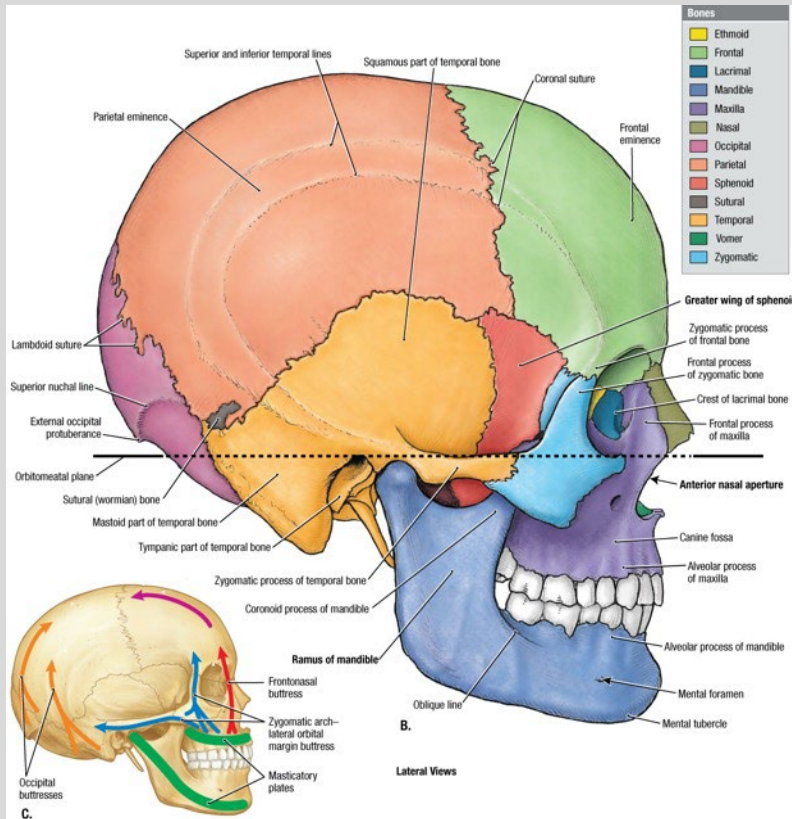
The motion pattern of the cranium may also be altered into a strain pattern by a variety of other factors.

- Fascial pull from below
- Local cranial dysfunction
- Spinal and/or pelvic asymmetry or dysfunction

# Case #1

- After a thorough explanation the mom agrees to treatment with OCMM.
- You place your hands on the head for the vault hold. The mom ask you what are you feeling for?

# The Vault



- Virtually all hand contacts are on the vault, or membranous bone. From these contacts, the ultimate goal is to perceive what is and is not moving in the base.

# Characteristics of Strain Patterns

- Axis (or Axes) of Rotation - where are they located and orientation
- Motion - Direction of Rotation of the Sphenoid and Occiput on the axes
- Nomenclature - Generally named for the direction of motion of the basisphenoid relative to the basiocciput
- Morphology - General shape of cranium and regional characteristics
- Etiology – describe potential source for dysfunction
- Palpatory Diagnosis – to be able to described exaggerated motions through vault contact



# Cranial strain patterns at the SBS

Physiologic – generally adaptive and generally do not arrest flexion & extension

- Flexion (Inhalation)
- Extension (Exhalation)
- Torsion
- Sidebending and rotation

Non-Physiologic – generally maladaptive, restrict motion and disrupts the midline at the SBS

- Vertical Shears
- Lateral Shears
- Compression



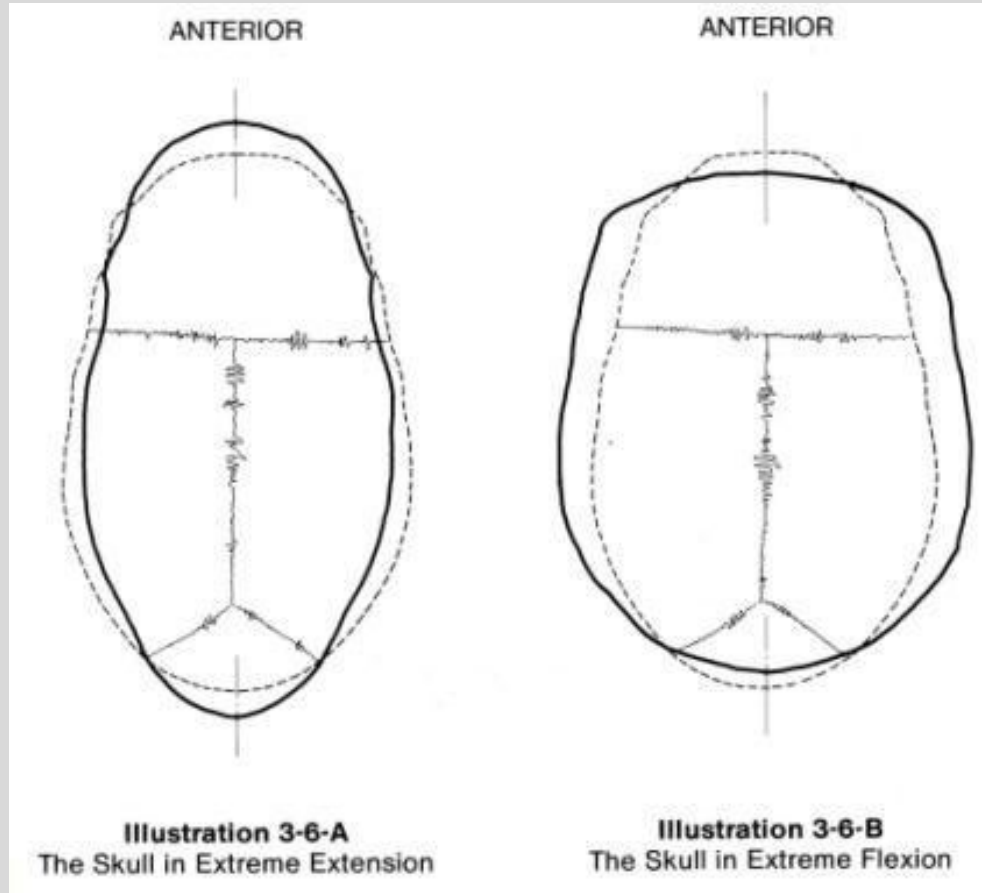
# Flexion / Extension Strain Patterns

- Flexion and extension occurs about **two parallel transverse axes**.
  - Sphenoid – through the body of the sphenoid at the level of the SS pivot
  - Occiput - above jugular processes through foramen magnum
- The sphenoid and occiput rotate in opposite directions on their axes.
- Physiologic motion





# Flexion or Extension Cranial Strain Patterns



- Like the spine, if either flexion or extension becomes exaggerated, with motion in the opposite direction restricted, this is considered dysfunction.

# Flexion of the SBS

- Occurs about 2 transverse axes
- Sphenoid rotates anteriorly and the occiput rotates posteriorly
- SBS rises with flexion
- External rotation of the paired vault bones
- Increased transverse diameter of the head
- Decreased AP & vertical diameter of the head



# Extension of the SBS

- 2 transverse axes
- Sphenoid rotates posteriorly and the occiput rotates anteriorly
- SBS drops with extension
- Internal rotation of the paired vault bones
- Decreased transverse diameter of the head
- Increased AP & vertical diameter of the head



# Flexion & Extension in Vault Contact

- Flexion – the greater wings of the sphenoid and occipital squama move inferiorly as SBS rises. Unpaired midline bones moves into flexion while paired bones move into external rotation (the operator's fingers and palms sense transverse swelling as they are moved towards the feet and laterally)
- Extension - the greater wings of the sphenoid and occipital squama move superiorly as SBS falls. Unpaired bones moves into extension while paired bones move into internal rotation (with receding of the PRM, the operator's fingers come together medially and move towards the vertex)



Flexion



Neutral



Extension

# Case #1 continued

- As you are palpating the head you feel the fingers of your right hand spread out more and are being pulled down to the right inferiorly towards the feet.
- Which strain pattern are you feeling?
- What may be contributing to the strain pattern?
- What visual findings would you expect with this pattern?



# Case #1 Continued

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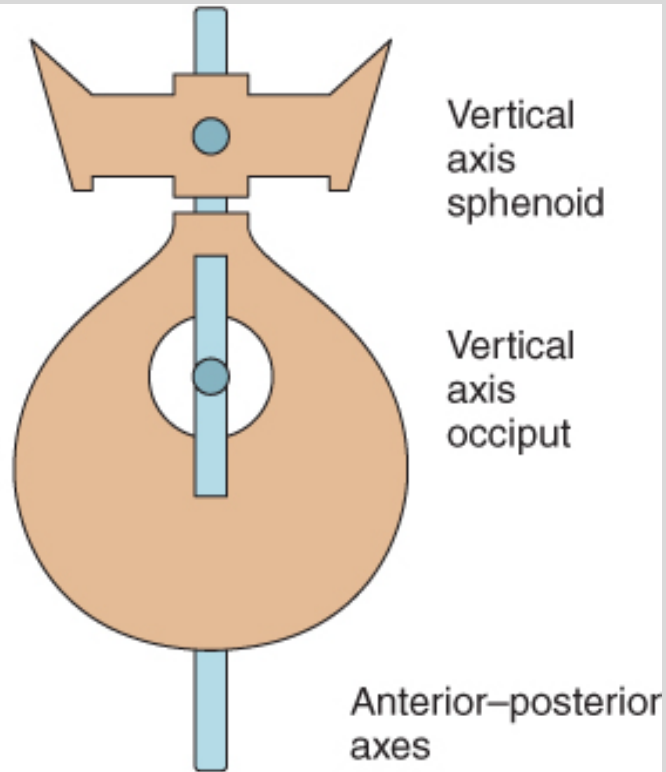
College of Osteopathic  
Medicine





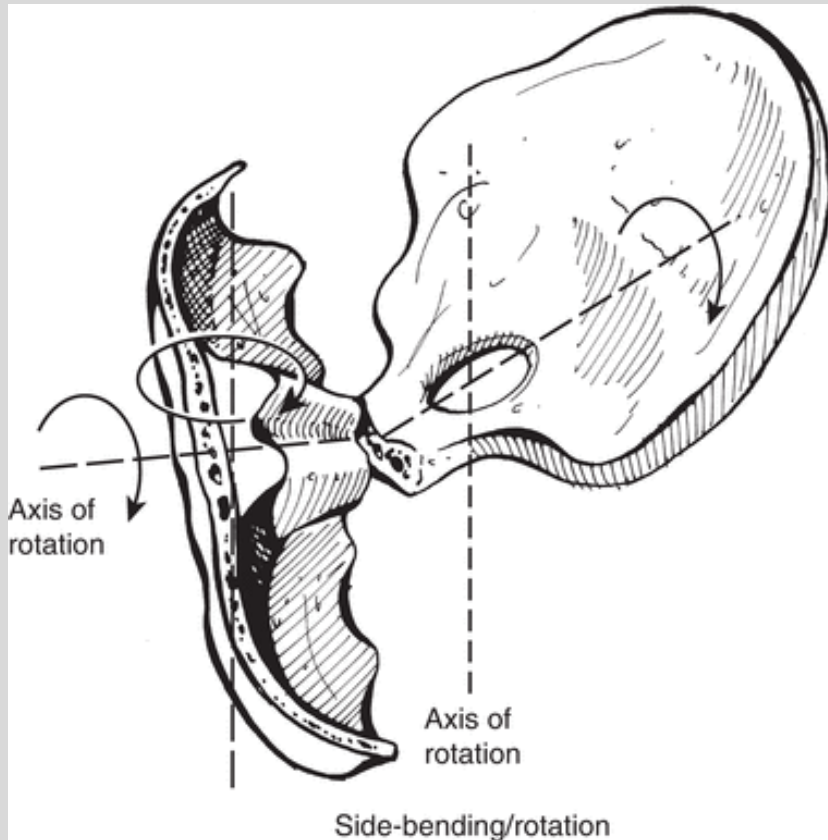
# Sidebending/Rotation

Superior view



- 3 Axes in total
- Sidebending occurs when the sphenoid and occiput move in opposite directions around 2 parallel vertical axes, one through the foramen magnum of occiput and the other through the body of sphenoid.
- Rotation occurs as both sphenoid and occiput rotate in the same direction around an A-P axis.
- They rotate to the side of the convexity which bringing that side more inferior.

# Sidebending/Rotation

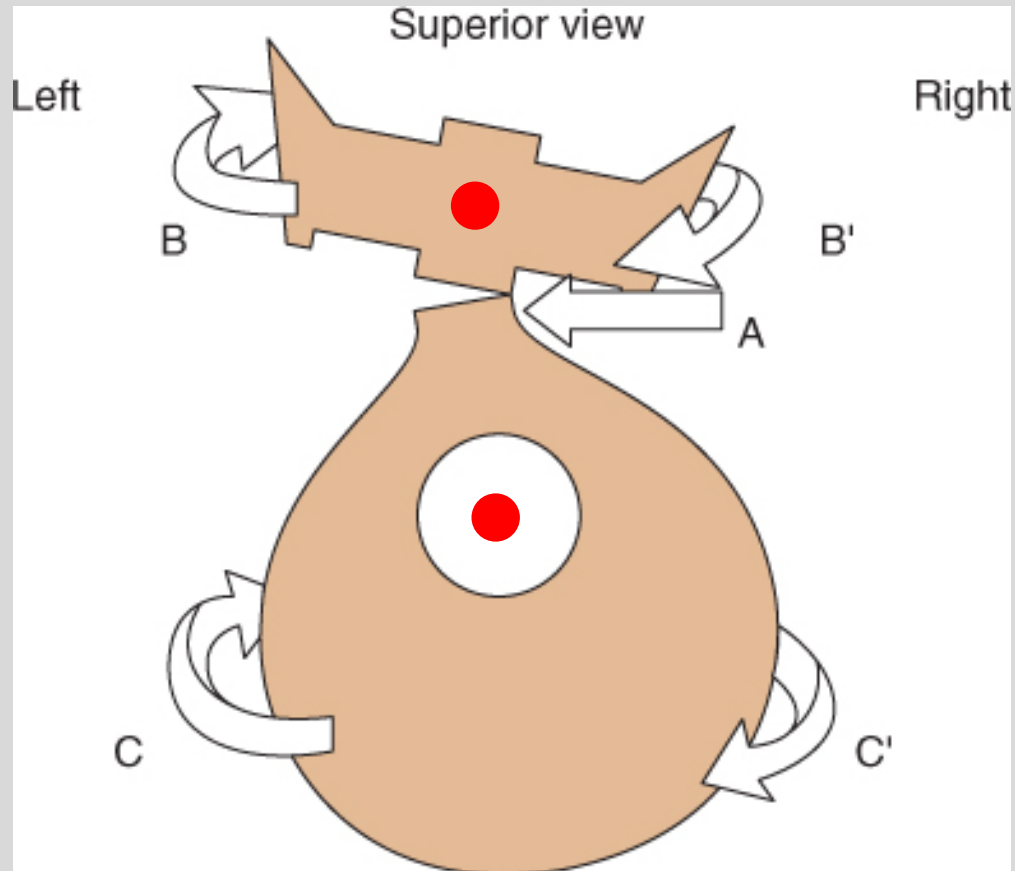


- The rotation occurs around the same A-P axis as torsion, however now the occiput and sphenoid rotate in the same direction
- The sidebending rotation is named for the side of the convexity similar to naming of scoliosis in the spine (also = direction of the rotation).

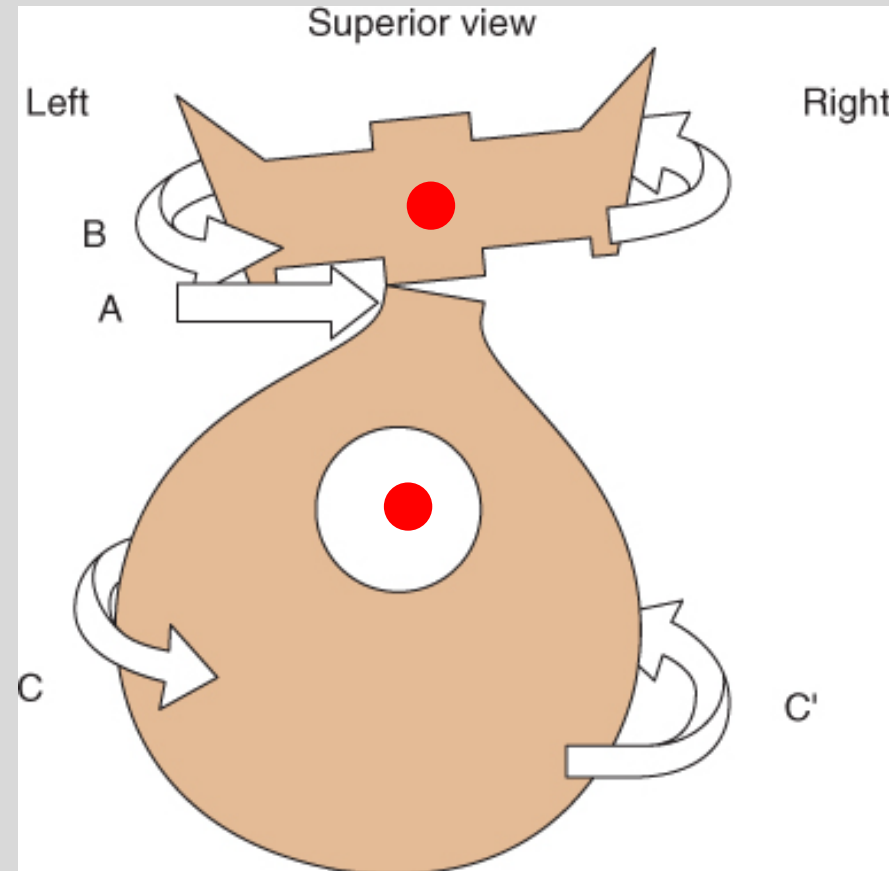
Left SB and Rotation

# Rotation about 2 vertical axes of the occiput and sphenoid in OPPOSITE direction ->sidebending

- Left SB-Rot

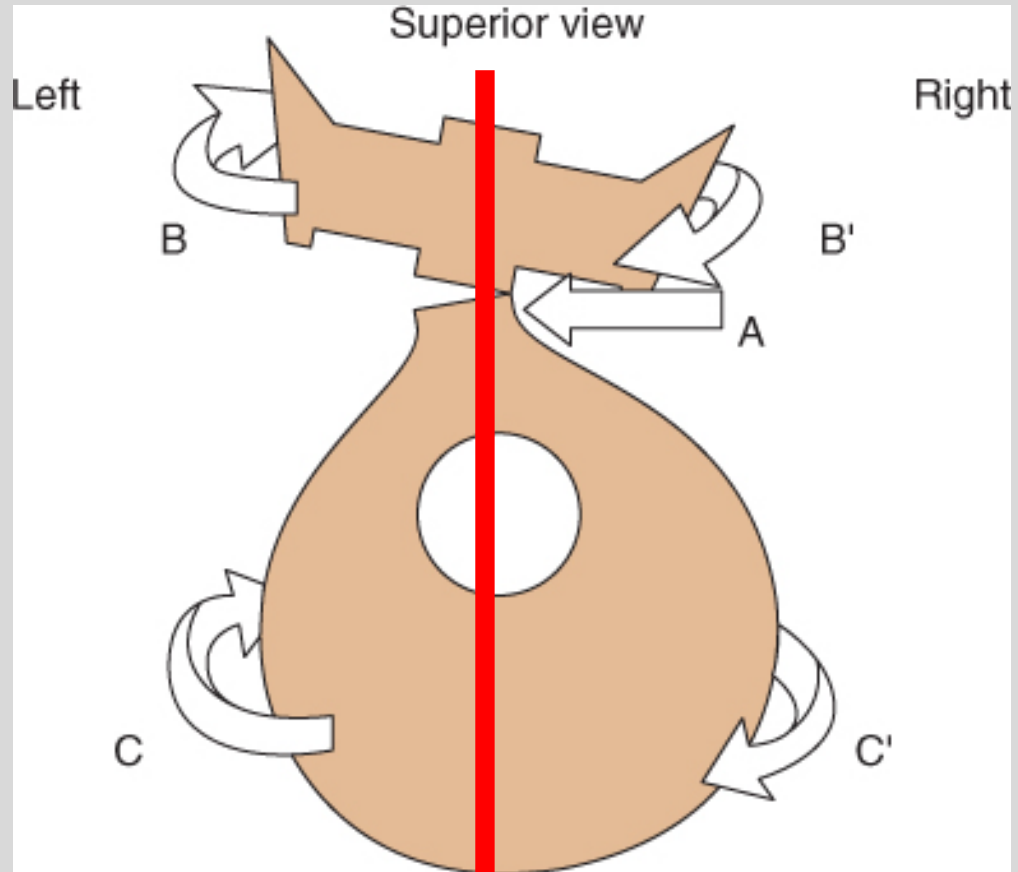


- Right SB-Rot

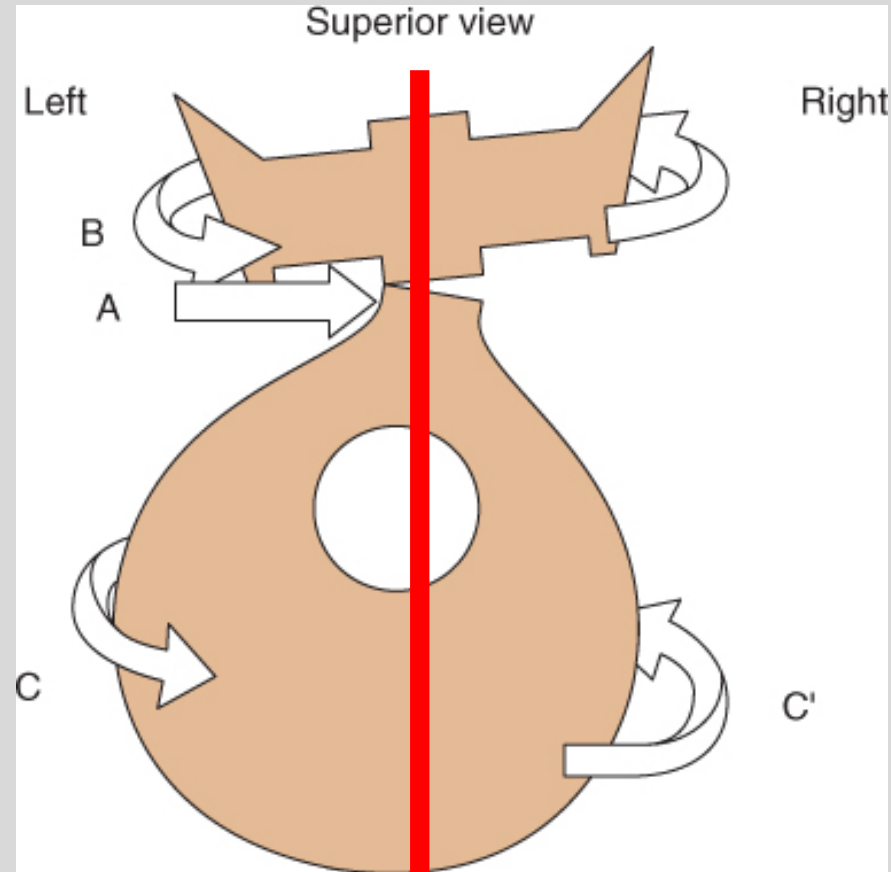


# Rotations of the occiput and sphenoid on an AP Axis in the SAME direction

- Left SB-Rot



- Right SB-Rot



# Sidebending/Rotation

## Right SB-Rot



## Left SB-Rot



- Vault Hold
  - Fingers move inferiorly and separate on the side of convexity (“DOWN and OUT”)
  - Fingers move superiorly and approximate on the side of concavity

Think of the side of the naming = convexity = think of cracking open the egg (sidebending) and then spilling out the contents (rotation)



# Visual Findings in sidebending rotation

TABLE 104-3

## Findings in Side-Bending and Rotation Dysfunctions of the Sphenobasilar Symphysis

| LANDMARK              | CONVEXITY     | CONCAVITY           |
|-----------------------|---------------|---------------------|
| Lateral frontal angle | Posterior     | Anterior            |
| Orbit                 | Narrow        | Wide                |
| Frontozygomatic angle | Lessened      | Increased           |
| Eyeball               | Retruded      | Protruded           |
| Zygomatic orbital rim | Prominent     | Flat                |
| Symphysis menti       | To this side  | Away from this side |
| Mastoid tip           | Posteromedial | Anterolateral       |
| Mastoid portion       | Anterolateral | Posteromedial       |
| Ear                   | Protruding    | Close to head       |

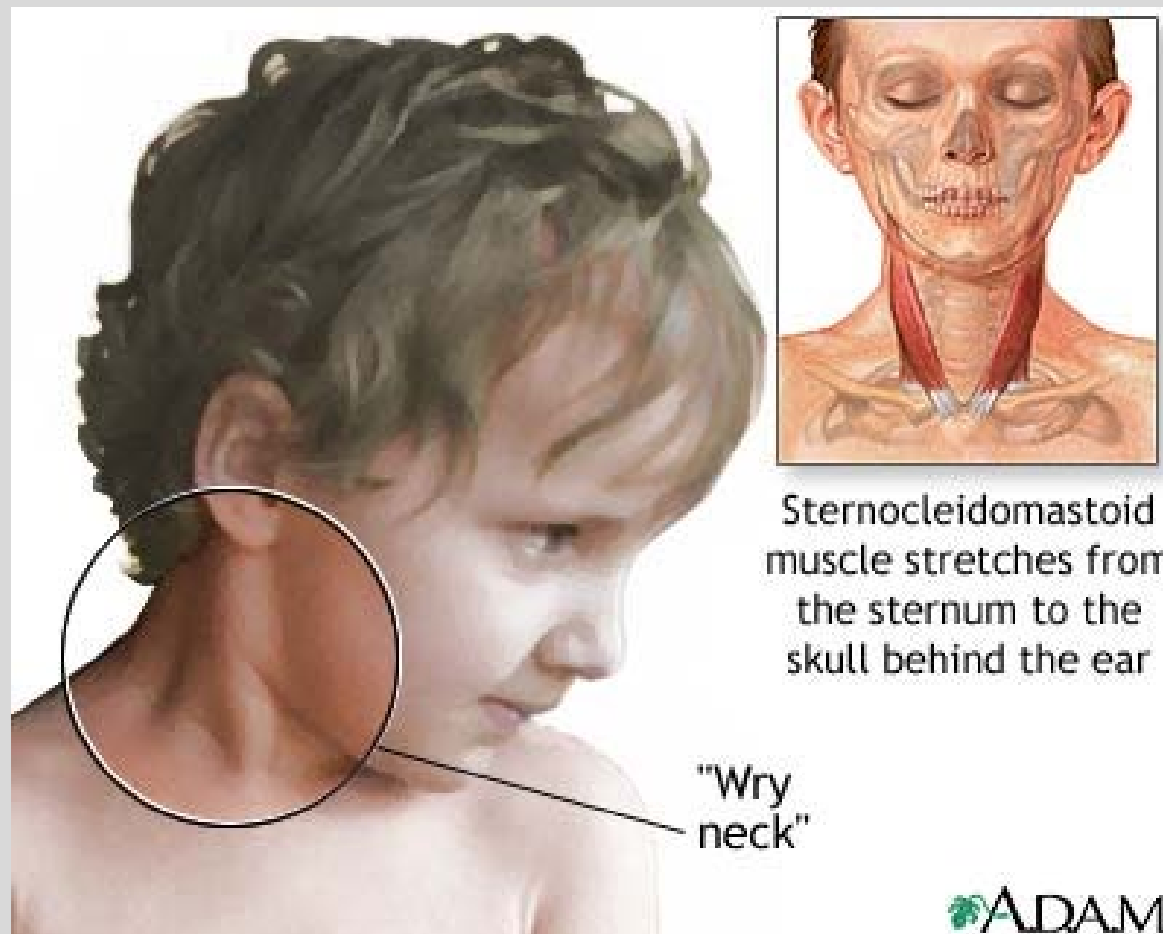


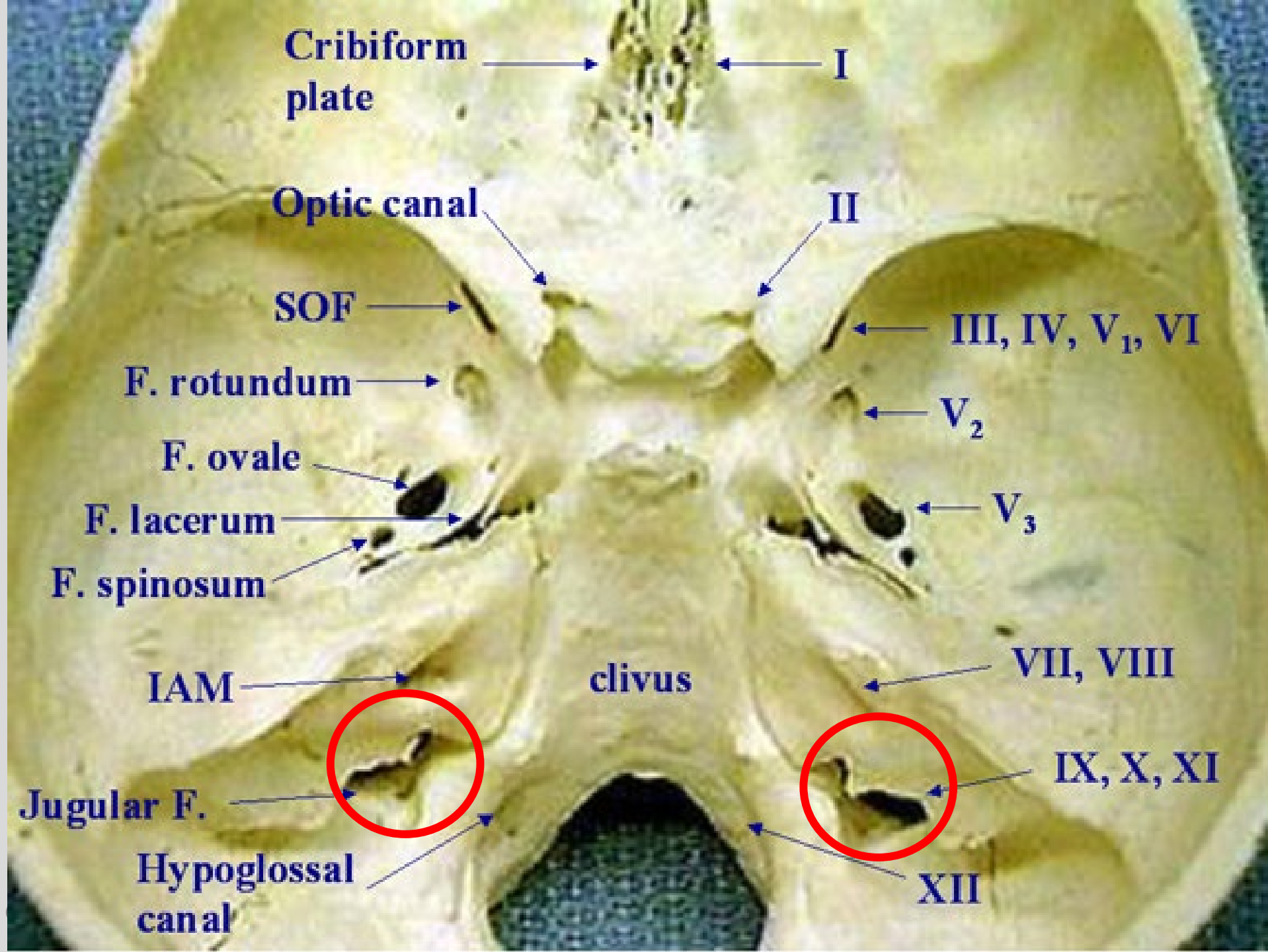


# Case #1 Continued

- You notice that the baby has decreased rotation to the right. What muscle can be involved with the case?
- What cranial nerve is associated with it?
- Where does that cranial nerve exit the skull?
- What other cranial nerves exits with that nerve and what bodily functions can be affected by irritation of those nerves?

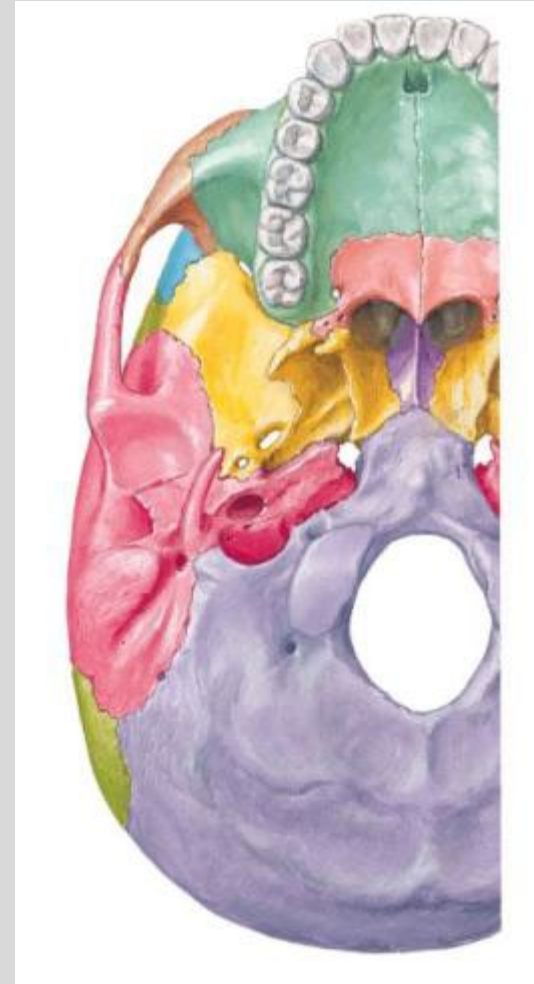
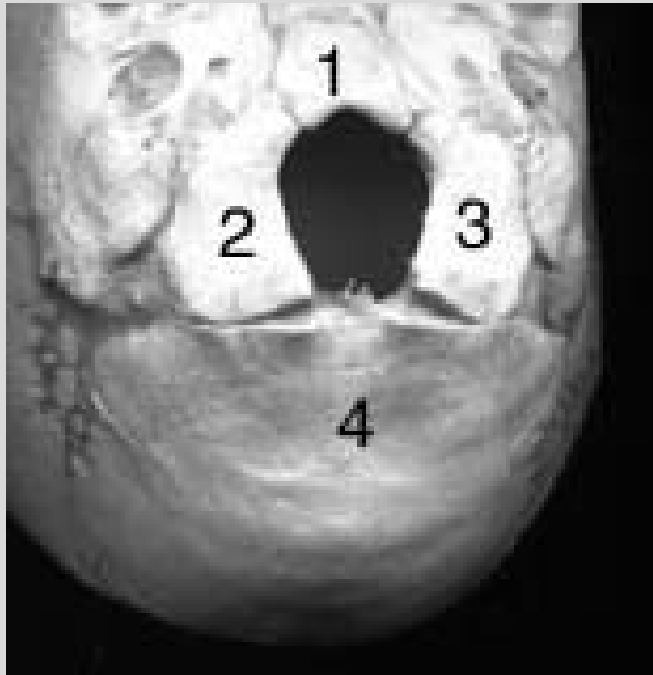
# Torticollis – Spinal Accessory – CN XI





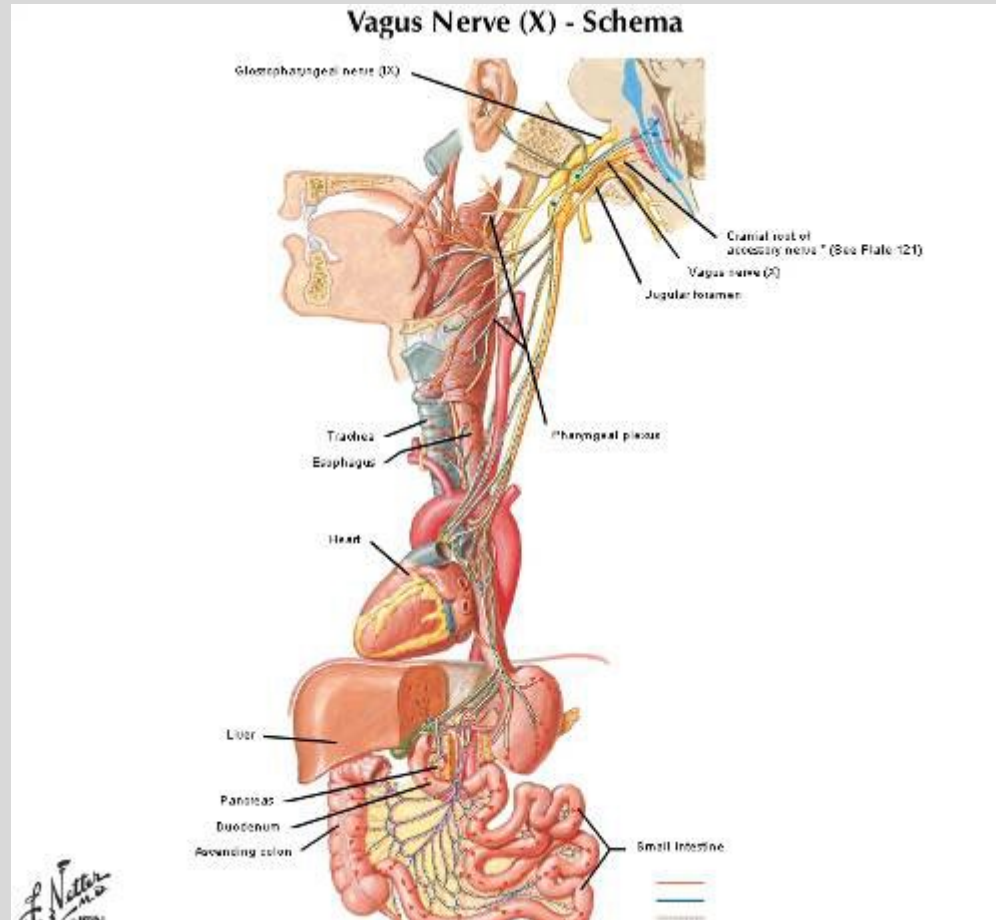
# Feeding problems / Swallowing

- CN 9 /10 – Jugular foramen
- CN 12 – Hypoglossal Canal

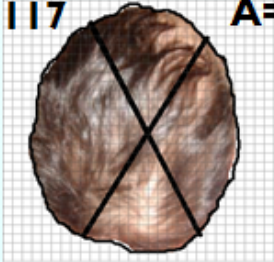
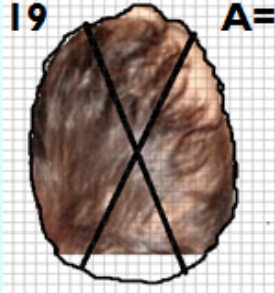


# Colic / Gastric irritation

- Jugular Foramen
- Vagus – CN X



# Results after OMM treatment

| Visit # | 1° Endpoint: Clinical                                                                                   |                                                                                                            | 1° Endpoint: Objective                                                                                                                           | 2° Endpoint:                                             |
|---------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
|         | Head Asymmetry                                                                                          | SCM                                                                                                        | CVAI /GRADE                                                                                                                                      | Motor Skills                                             |
| 1       |                        |                           | <p>B= 117      A= 128</p>  <p>CVAI = 9% Grade IV (Severe)</p> | 4-5 month level<br>(State Eval)                          |
| 2       | Clinical Improvement noted                                                                              | Mild increase in head rotation;<br>Persisting SCM spasms                                                   | Not Measured                                                                                                                                     | No change                                                |
| 3       | <br>Improvement noted | <br>Decreased SCM Spasms | <p>B= 119      A= 127</p>  <p>CVAI = 6% Grade II (Mild)</p>  | Moderate gain in motor skills: turns over independently; |



# Case #1 continued

- You were able to help with the torticollis after one visit but the baby's head is still misshaped. Looking picture below which strain pattern is still evident?
- What is the medical terminology for a misshaped head?
- How can OMT help with that diagnosis?



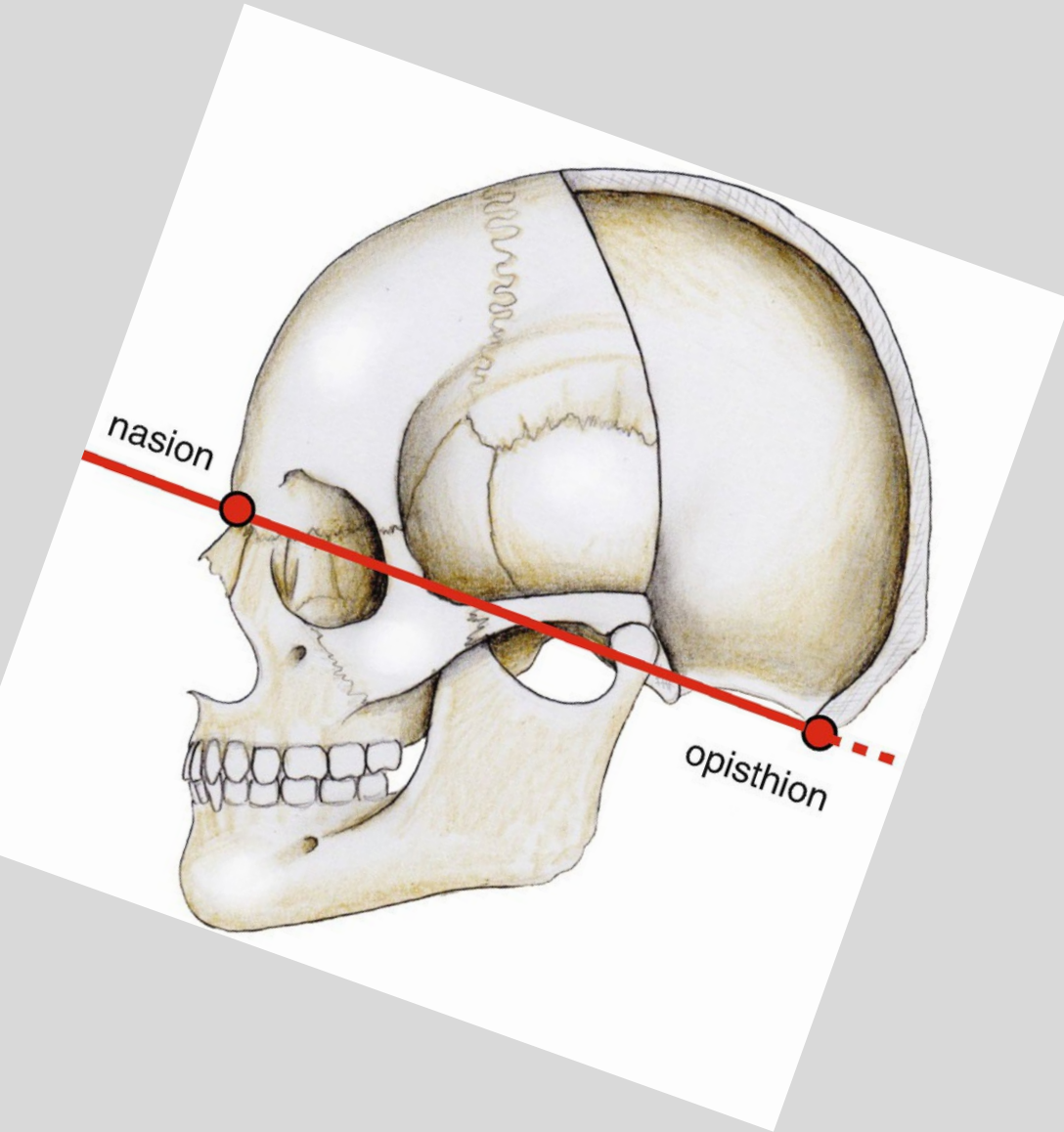
# Visual findings in torsion dysfunctions

**TABLE 104-2 FINDINGS IN TORSION  
DYSFUNCTION OF THE SPHENOBASILAR  
SYMPHYSIS**

| PARAMETER<br>EVALUATED   | SIDE OF HIGH<br>SPHENOID<br>WING     | SIDE OF LOW<br>SPHENOID<br>WING       |
|--------------------------|--------------------------------------|---------------------------------------|
| Frontal lateral<br>angle | Anterior                             | Posterior                             |
| Orbit                    | Wide                                 | Narrow                                |
| Frontozygomatic<br>angle | Increased                            | Lessened                              |
| Eyeball                  | Protruded                            | Retruded                              |
| Zygomatic orbital<br>rim | Everted and<br>externally<br>rotated | Inverted and<br>internally<br>rotated |
| Symphysis menti          | To this side                         | Away from<br>this side                |
| Mastoid tip              | Posteromedial                        | Anterolateral                         |
| Ear                      | Protruding                           | Close to head                         |



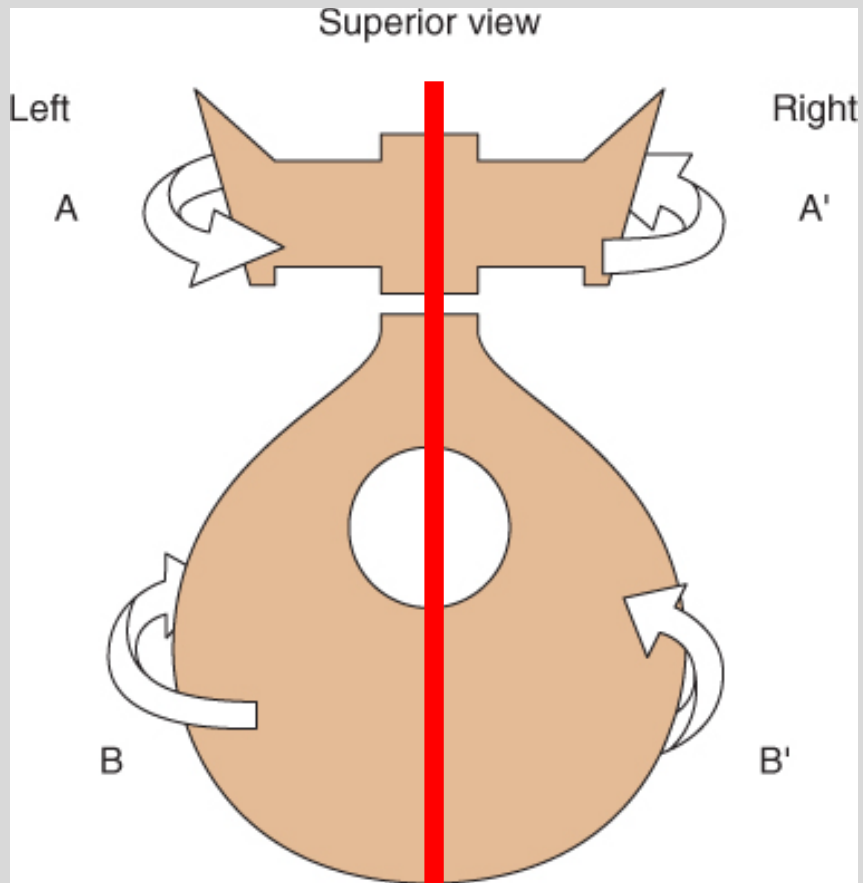
# Torsion Strain Pattern



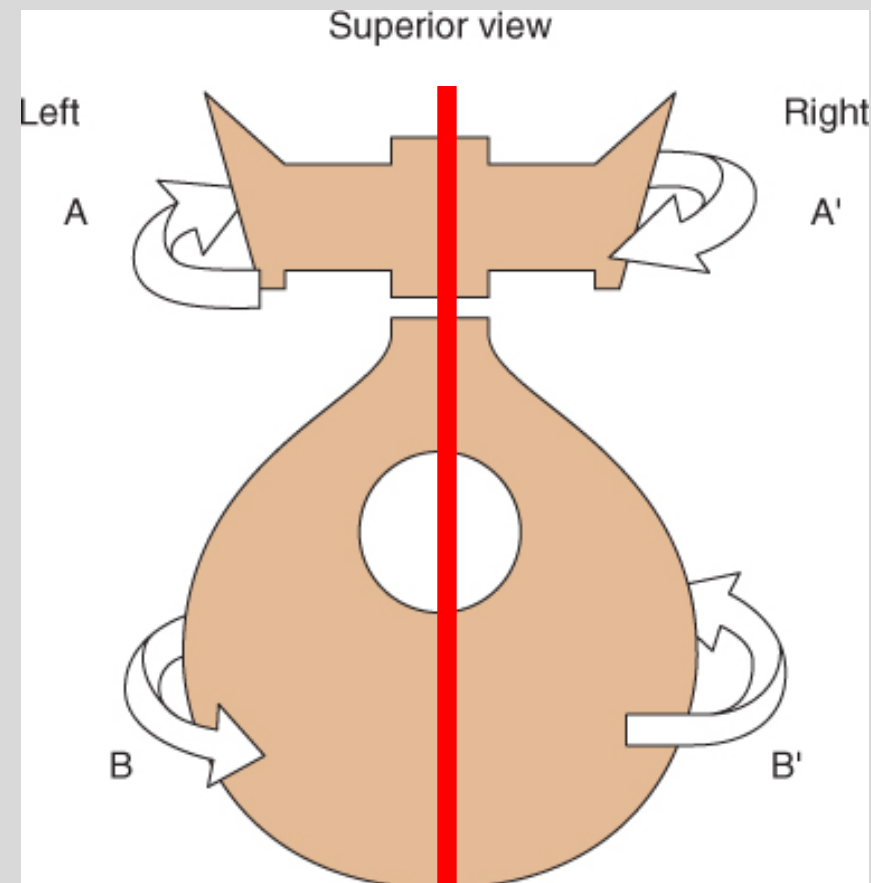
- One anteroposterior axis, from nasion to opisthion
- Sphenoid and occiput rotate in opposite directions about the A/P axis
- Physiologic strain pattern
- Torsion is named for the high wing of the sphenoid

# Torsion Strain Pattern

## Left Torsion



## Right Torsion



# HOW TO DEMO STRAINS - Torsion:

## Right Torsion



## Left Torsion



- Vault Contact
  - Hands rotate in opposite directions (pictures obvious exaggerations)
  - The index finger on the greater wing of the sphenoid moves cephalad on the side of torsion



# Plagiocephaly / Brachycephaly



- **Plagiocephaly** - condition characterized by an asymmetrical distortion (flattening of one side) of the skull
- **Brachycephaly**, also known as flat-head syndrome - shortened front-to-back diameter of the skull.





- American Academy of Pediatrics (AAP) released its recommendation in 1992 that infants be placed down for sleep in a nonprone position.
- Helped to decrease rate of Sudden Infant Death Syndrome (SIDS) > 50%
- Increased incidence of plagiocephaly



# Treatment Options

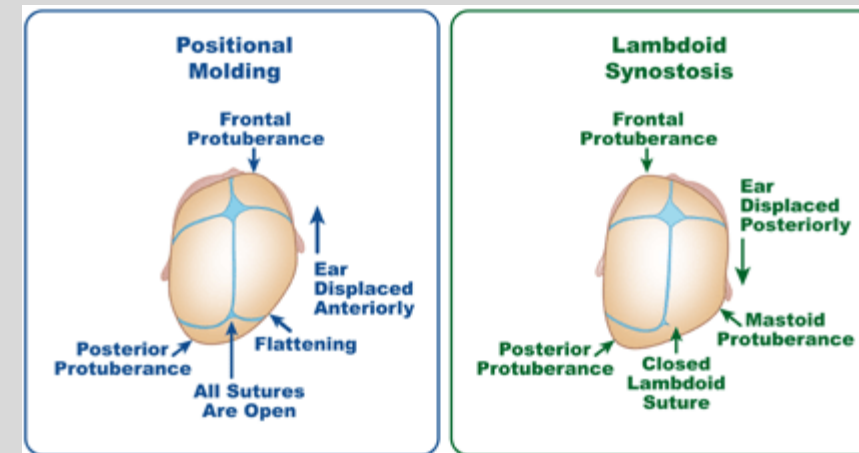
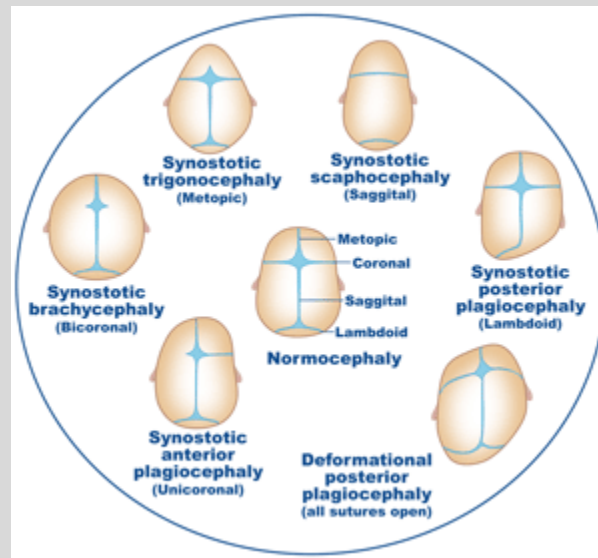
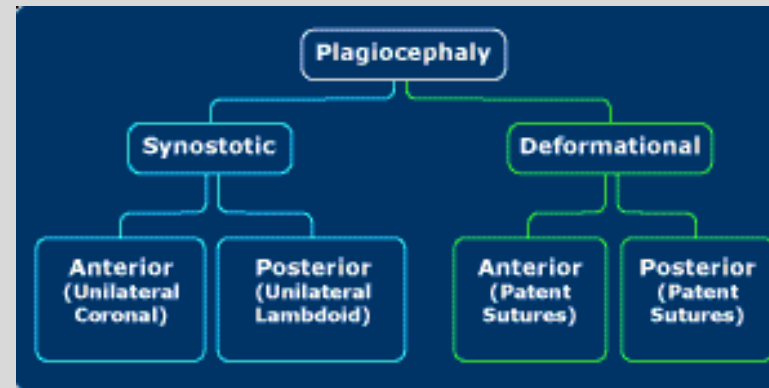


Benjamin treating Lydia

VS



# Craniosynostosis



# Case # 2

- A 6-year-old girl presents with problems with her vision and decreased abduction of her right eye when asked to look to the right since birth.



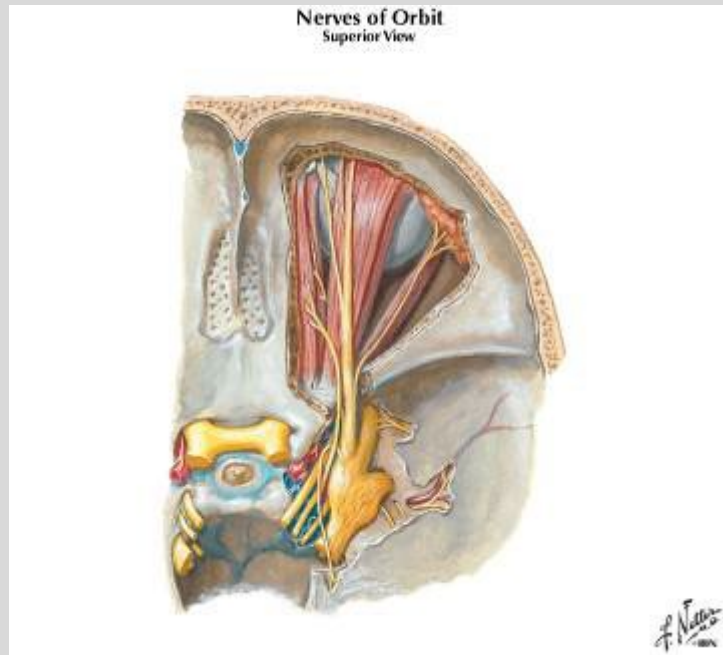
# Questions?

- What is her diagnosis?
- What cranial nerves are affected and how do they exit the skull?
- What type of strain pattern could have affect the cranial nerves?

# Extra-ocular muscle disorders

## Strabismus / Nystagmus

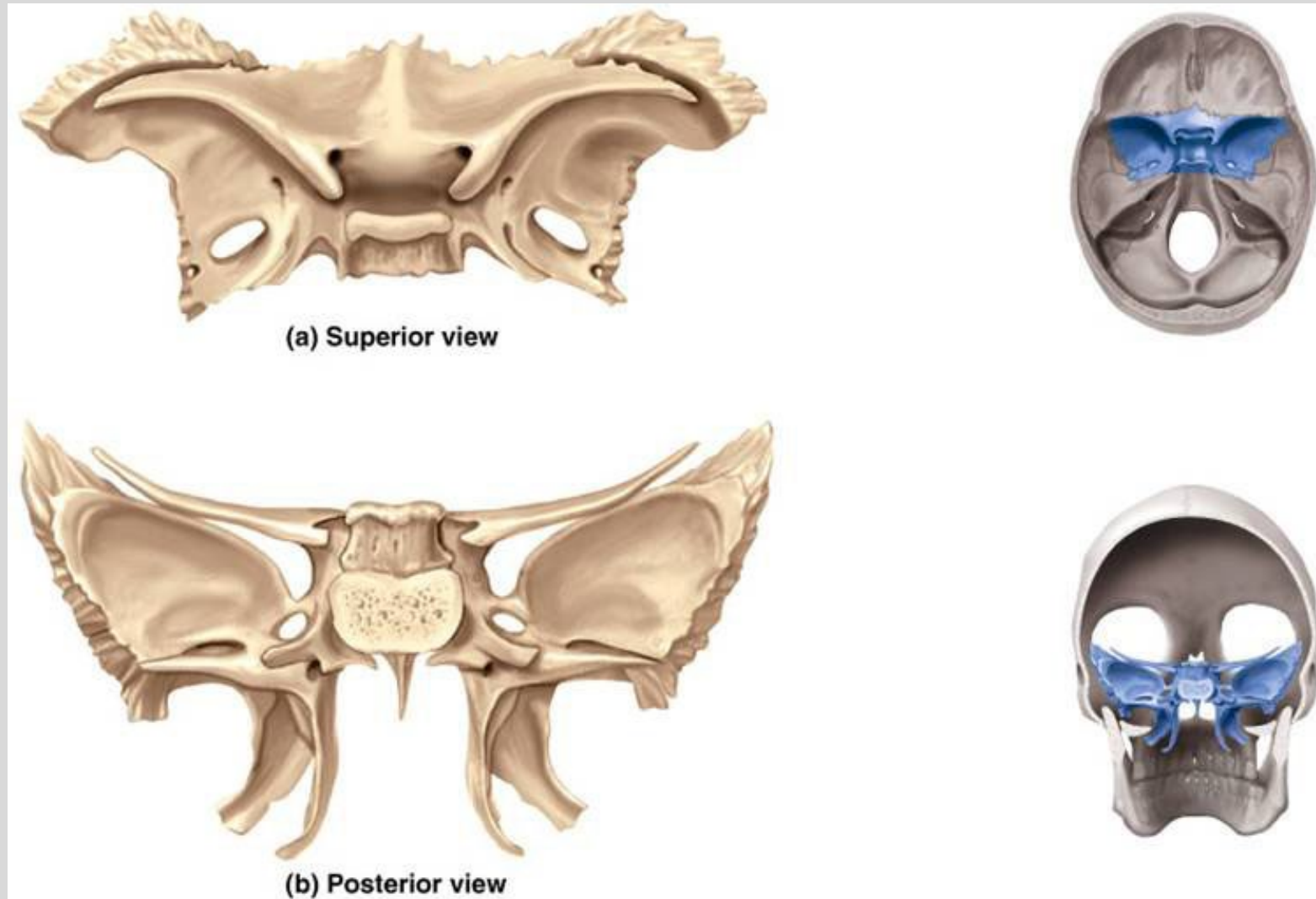
CN 3,4,6 pass through the superior orbital fissure of the sphenoid.



- CN 3 - oculomotor nerve - eye's movements & constriction of the pupil
- CN 4 - trochlear nerve - superior oblique muscle
- CN 6 - abducens Nerve - lateral rectus muscle



# Sphenoid



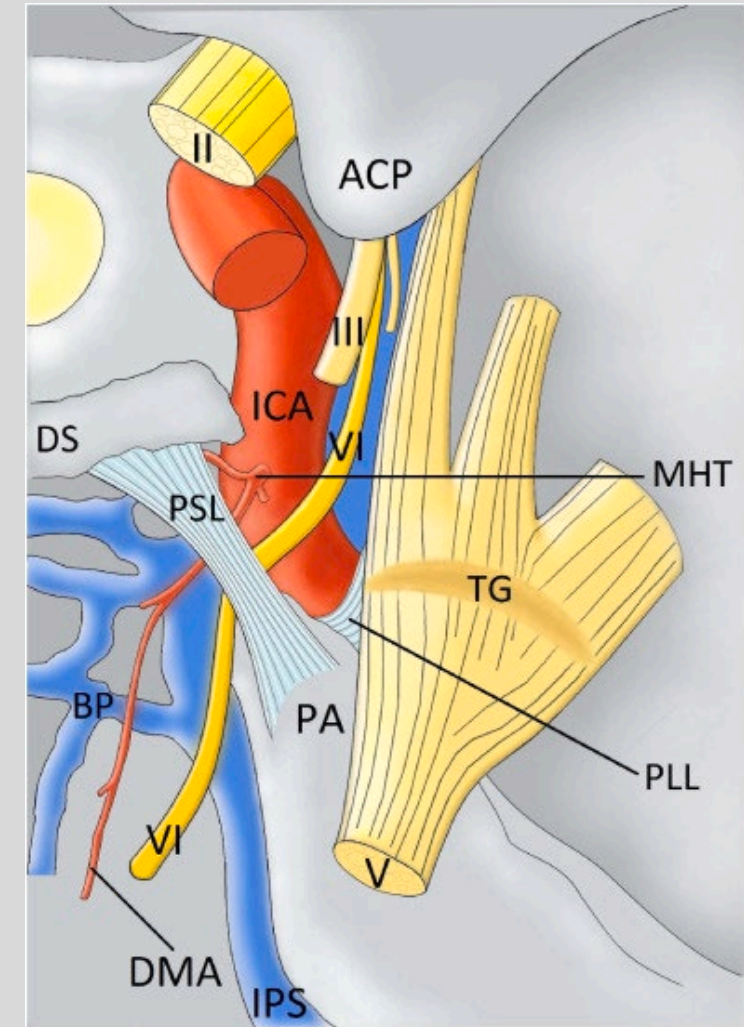
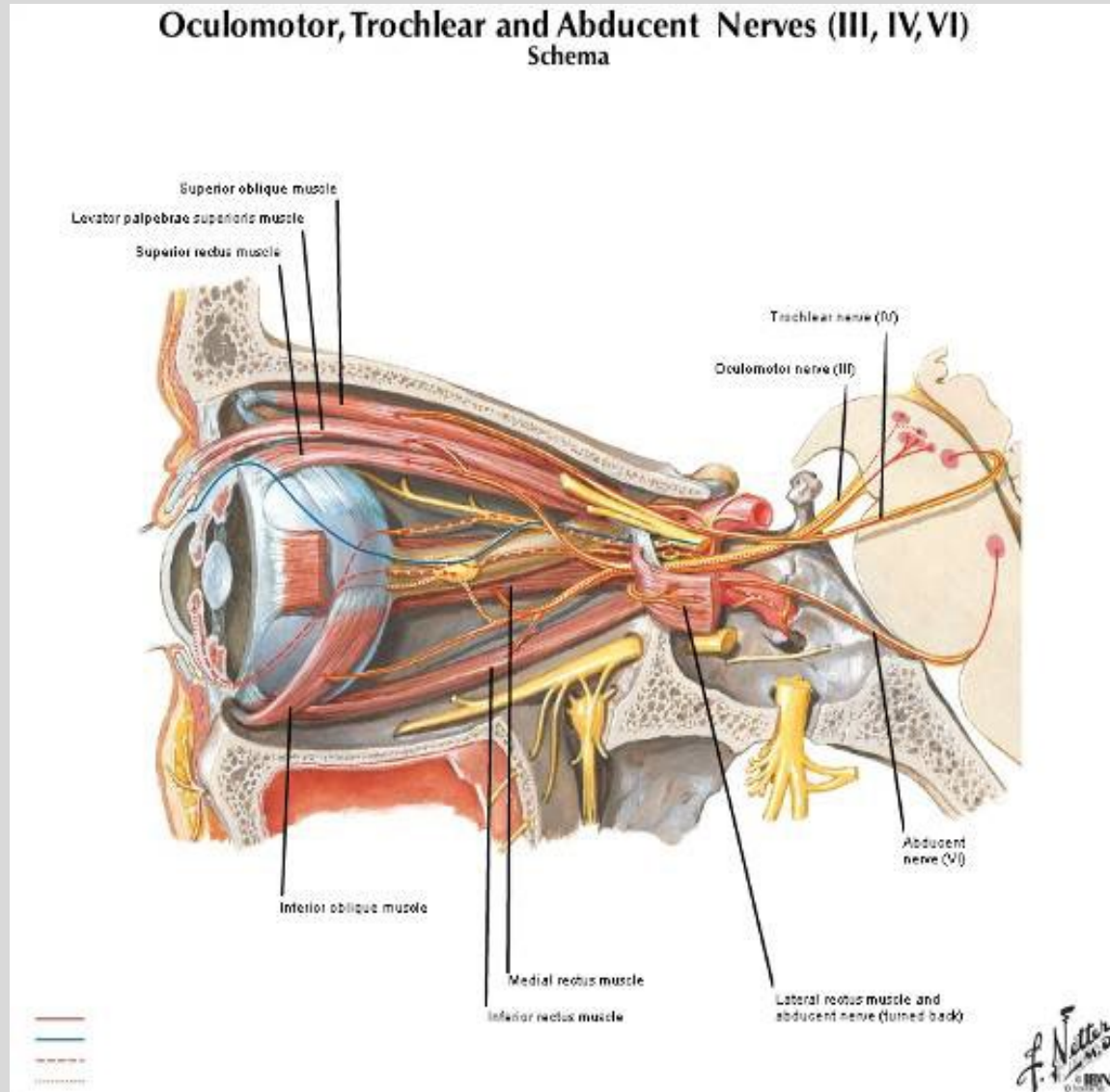
# Duane's Retraction Syndrome (DRS)

- Congenital absence or poor development of the abducens nuclei, with aberrant innervation by the oculomotor nerve, that leads to impaired ocular motility.
- 80% of these cases are unilateral and patients have limited abduction, adduction or both.
- Amblyopia occurs in 3 to 25% of patients due to strabismus rather than a refractive difference

# Duane's Retraction Syndrome

- While DRS is characterized by a miswiring of the lateral and medial recti muscles, a similar clinical picture may result from various mechanical, anatomical, and innervational disorders.
- The abducens nerve, responsible for abduction of the eye.
- Dural attachments cause the nerve to angulate as it passes under the petrosphenoidal ligament (Gruber's ligament).
- Due to this anatomic relationship, compression or stretching of the nerve may be the cause of various cranial nerve palsies.
- Any change or interruption to the wiring of the lateral rectus may compromise its function.
- Treatment for patients with DRS may include corrective lenses or eye muscle surgery to improve head position but it does not normalize horizontal motility.

# Abducens – CN VI Petrosphenoid ligament



# Case findings

| Region   | Structural Finding                                                                                                          |
|----------|-----------------------------------------------------------------------------------------------------------------------------|
| Cranial  | Right frontal compression<br>Right temporal external rotation<br>Right condylar compression<br>Right lateral strain pattern |
| Cervical | Right scalene muscle spasm ( <i>Fig. 4</i> )<br>Right paravertebral muscle spasm                                            |
| Thoracic | T1-2FSRR, right 1 <sup>st</sup> rib inhalation dysfunction,<br>T5ESRL, T9-12NSRRL                                           |



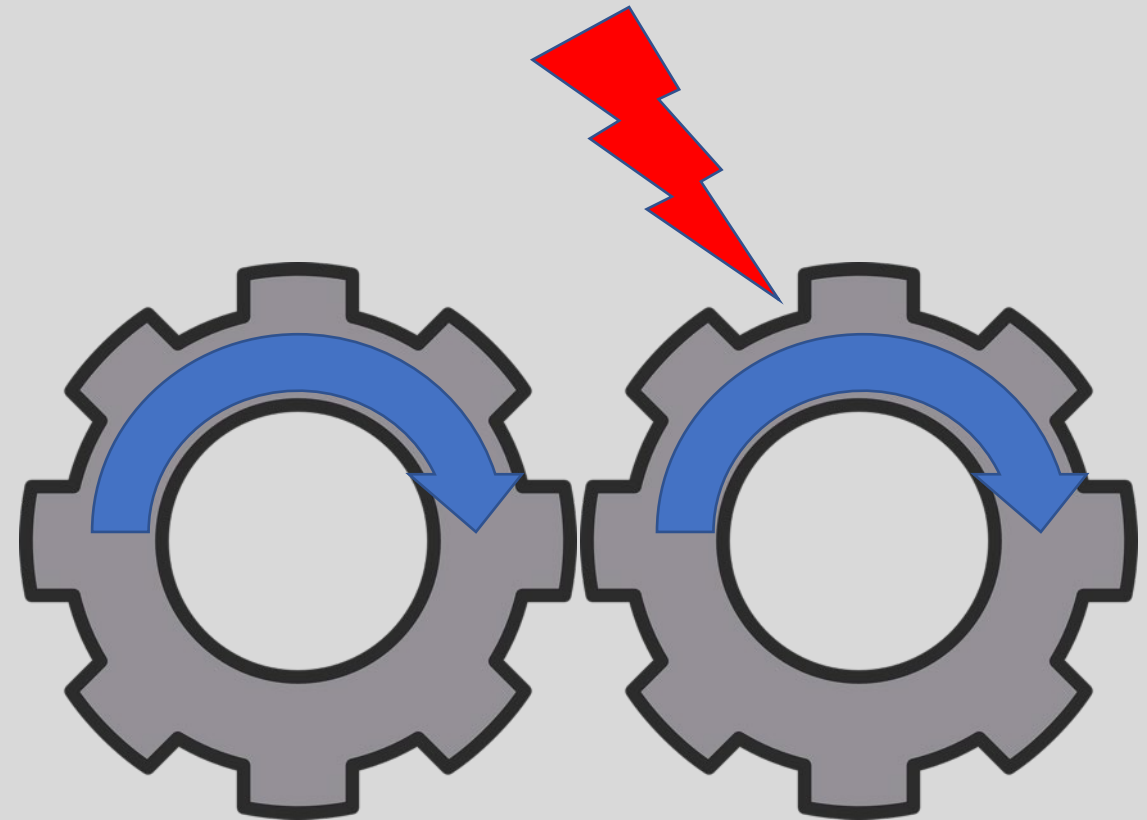
# SBS Cranial Strain patterns

## Physiologic

- Flexion (Inhalation)
- Extension (Exhalation)
- Torsion
- Sidebending and rotation

## Non-Physiologic

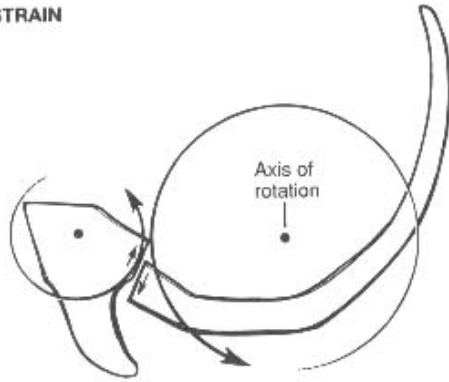
- Vertical Shears
- Lateral Shears
- Compression



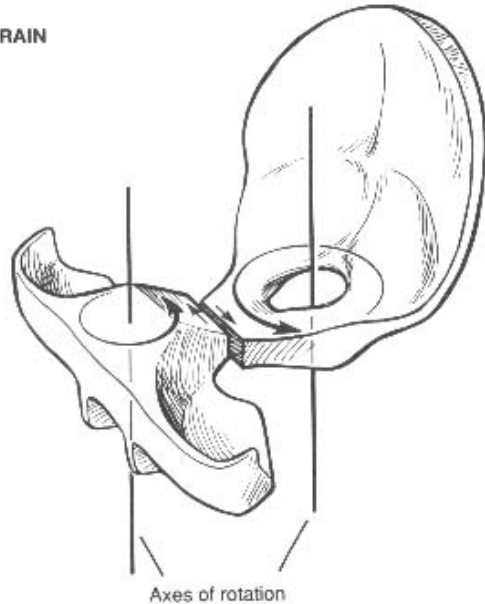


# Non-Physiologic Strain Patterns

VERTICAL STRAIN



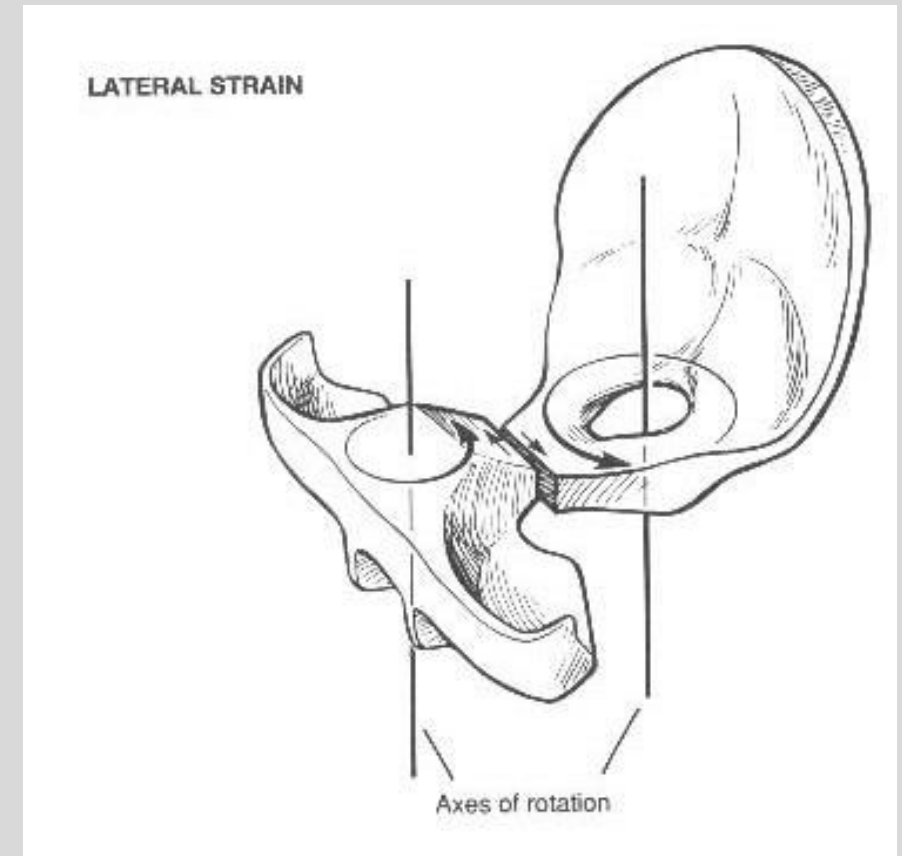
LATERAL STRAIN



- Vertical and lateral strains are non-physiologic, as they involve a shear at the SBS.
- Usually due to trauma that directly affects the SBS disrupting the midline.
- Typically symptomatic and interfered with normal motion thus having a pathologic effect.

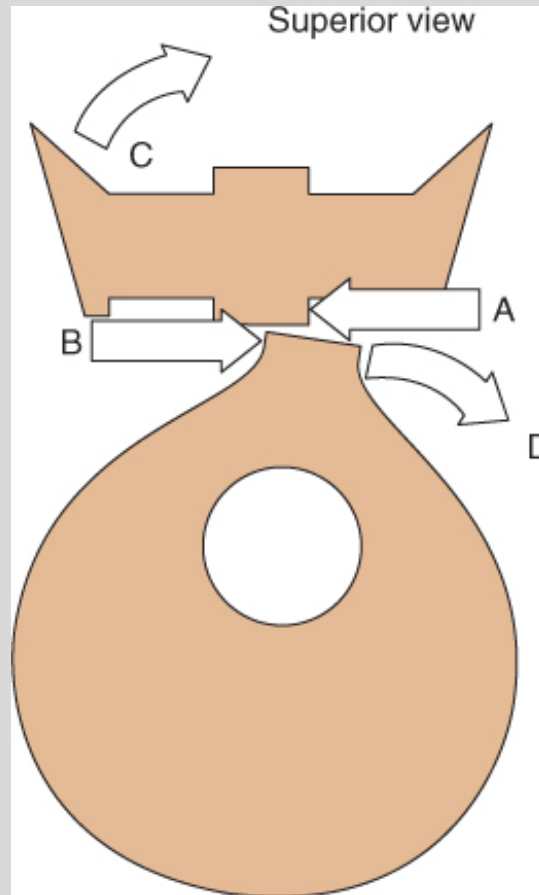
# Lateral Strains: “Parallelogram Head”

- Two parallel vertical axes
- The sphenoid and occiput rotate in the **same** direction.
- Lateral strains are named for the direction the basisphenoid moves towards.
- Non-physiologic strain may have varying and complex causes resulting in findings not exactly fitting the expected parallelogram.

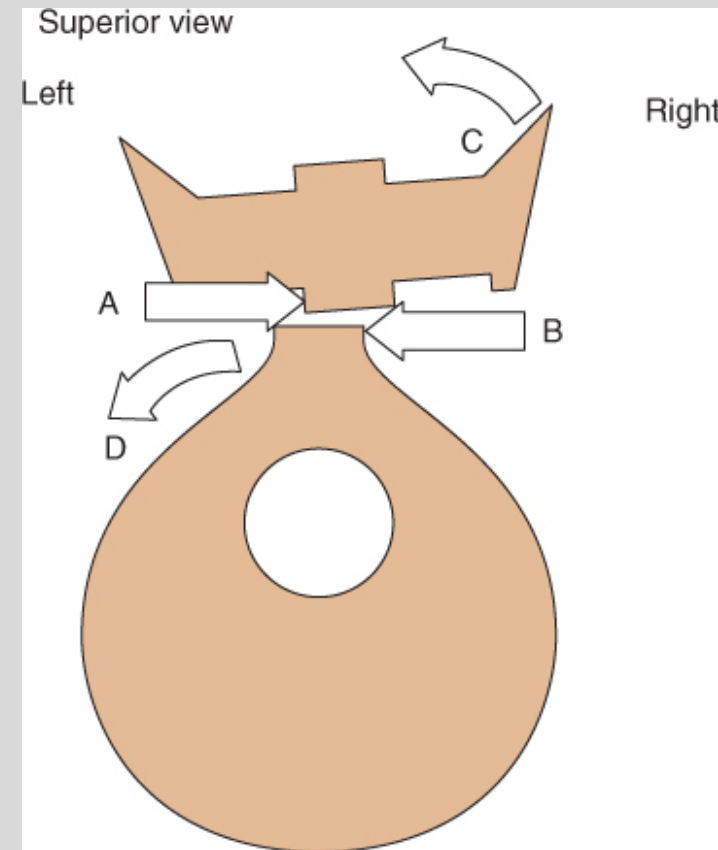




# Lateral Strains



Left Lateral Strain  
(feels like Greater Wing to  
the Right)

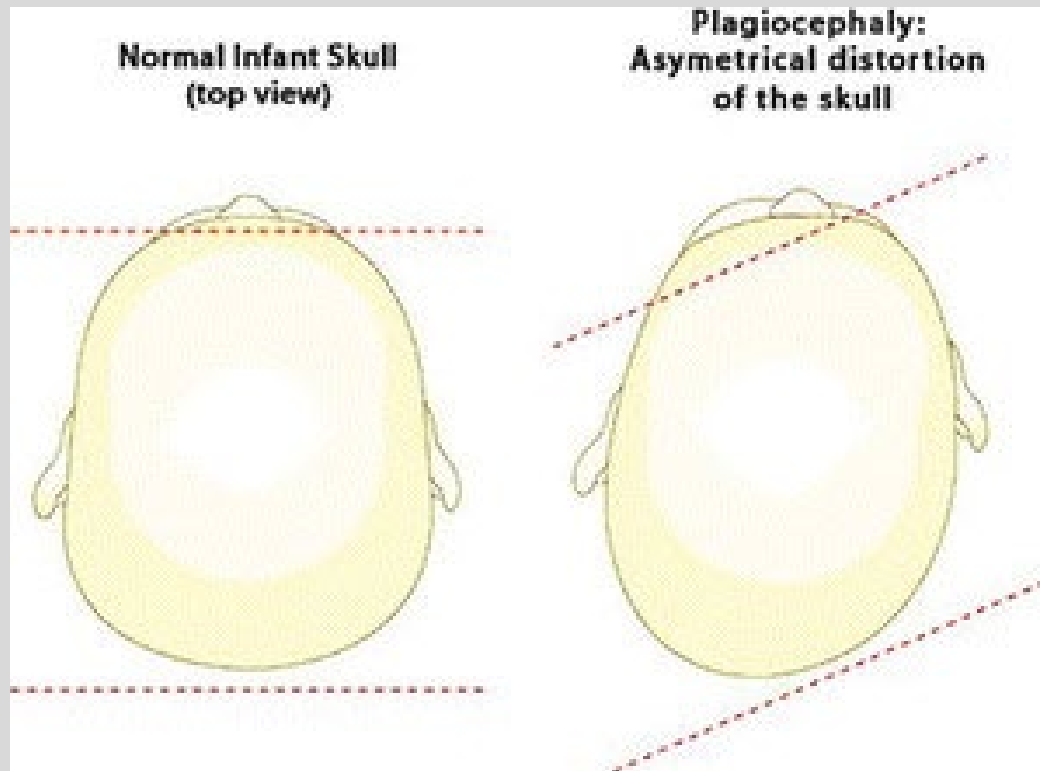


Right Lateral Strain  
(feels like Greater Wing to  
the Left)



# Lateral Strains: “Parallelogram Head”

- Can be due to compressive force, positional, in utero or plagiocephaly secondary to torticollis or trauma from a lateral force



# Lateral Strains

Left Lateral  
strain GW to  
the right



**Neutral**



Right Lateral  
strain GW to  
the left

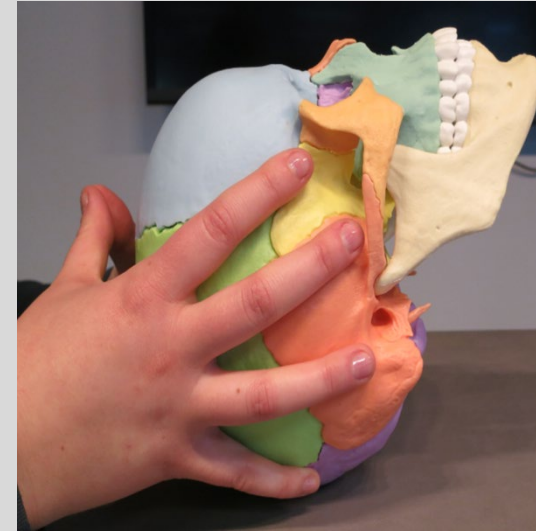


- Index fingers move to side the greater wings (GW), the pinkies move to the opposite side (generally the direction of the basi-sphenoid)
- Hands will form a parallelogram.

GW = Greater wing of the sphenoid

# Treatment & results

| Region   | Technique                      |
|----------|--------------------------------|
| Cranial  | Frontal lift                   |
|          | Temporal bone balancing        |
|          | Condylar decompression         |
|          | Balanced membranous tension    |
| Cervical | Articulation – Spine           |
|          | Facilitated positional release |
| Thoracic | Seated Articulatory            |
|          | Thoracic outlet BLT            |
|          | Muscle energy                  |



Pre-treatment

Post-OMT



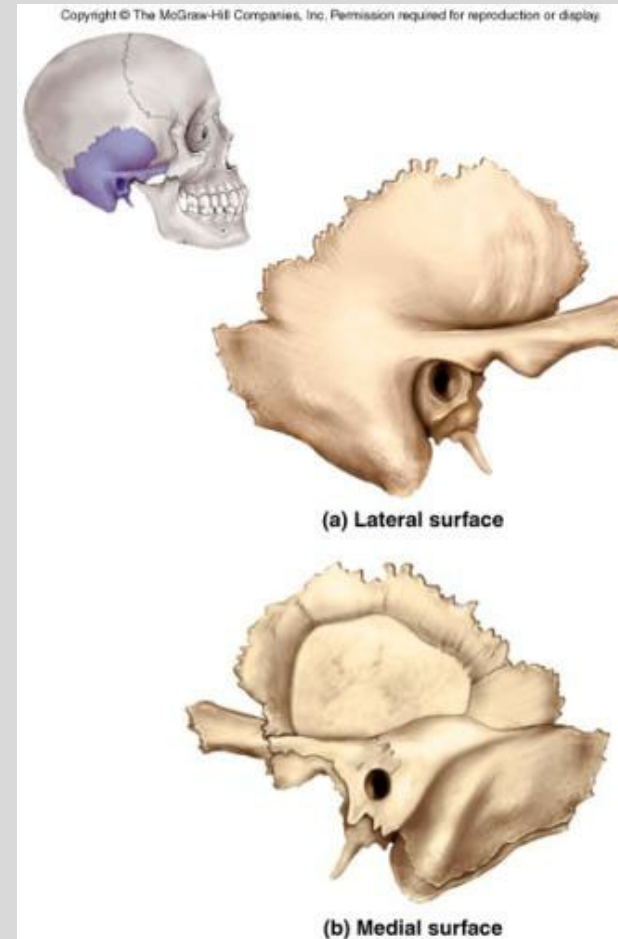


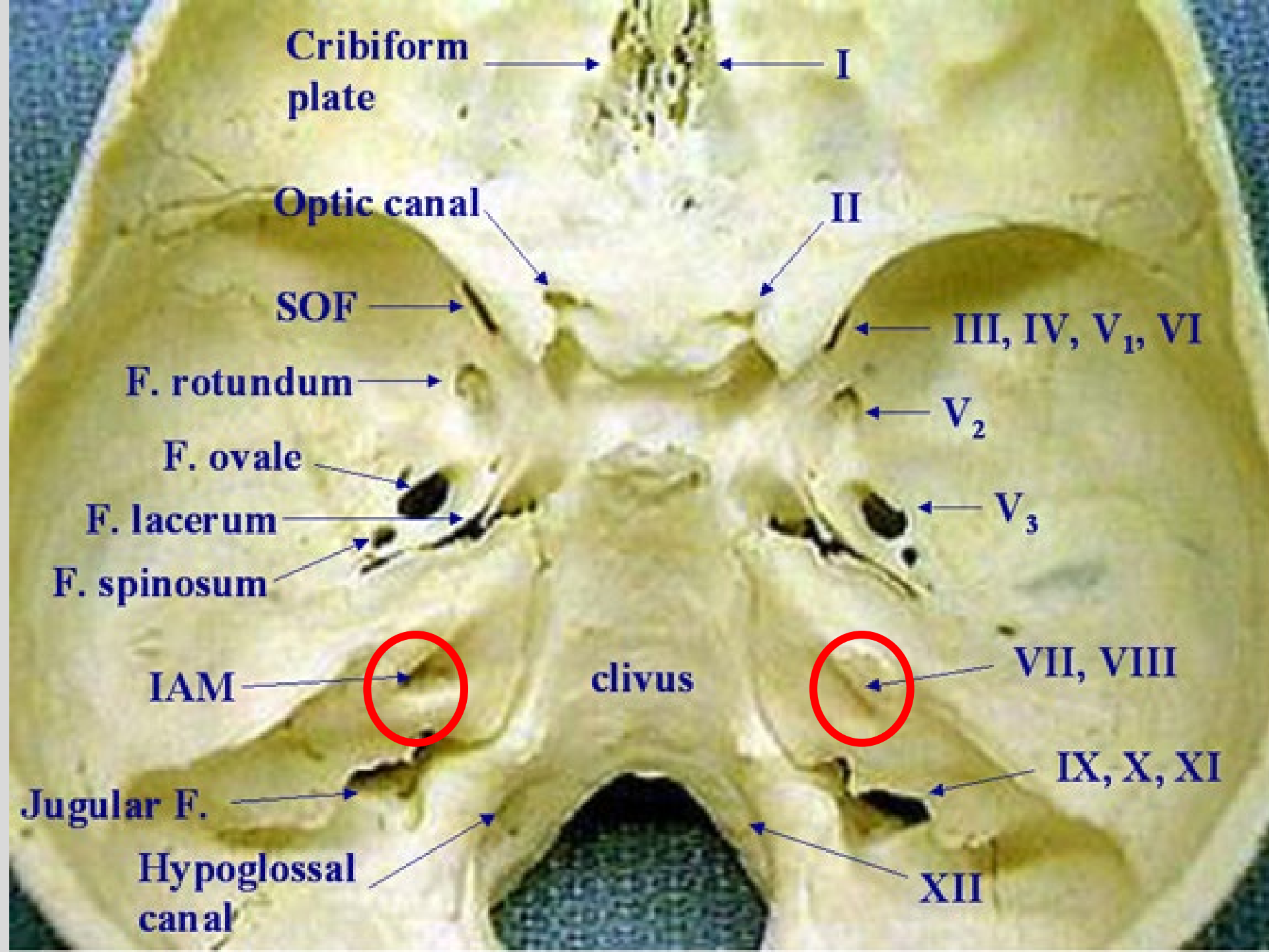
# Case #3

- A 50-year-old man presents with c/o dizziness, and a persistent ringing in his ear after tripping and hitting his chin on the dining room table as he fell to the ground.
- What cranial nerves are involved?
- What foramen does it exit from and what cranial bone is most involved?
- What strain pattern could have occurred in this case?

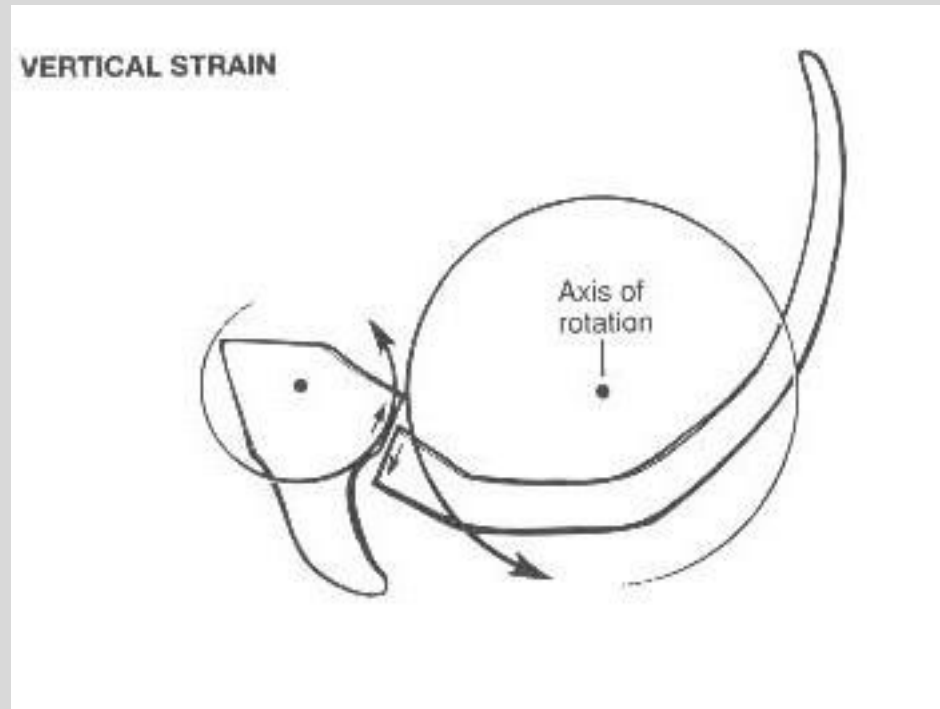
# CN VII & CN VIII – exits skull via the internal acoustic meatus

- CN VII - Facial Nerve
  - Bell's Palsy
  - PSNS innervation via Greater petrosal nerve sinuses
- CN VIII - sound and equilibrium
  - Tinnitus
  - Vertigo





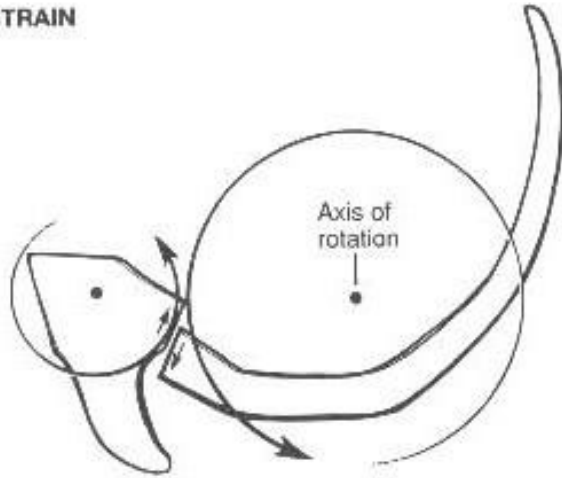
# Vertical Strains



- Two parallel transverse axes (same as for flexion and extension).
- However, the occiput and sphenoid are rotating in the same direction.
- Named for the position of the basisphenoid.
- Can be resultant from a blow to the head from above or from below.

# Vertical Strains

VERTICAL STRAIN



## Superior vertical strain

- Occiput rotates into relative extension
- Sphenoid rotates into relative flexion

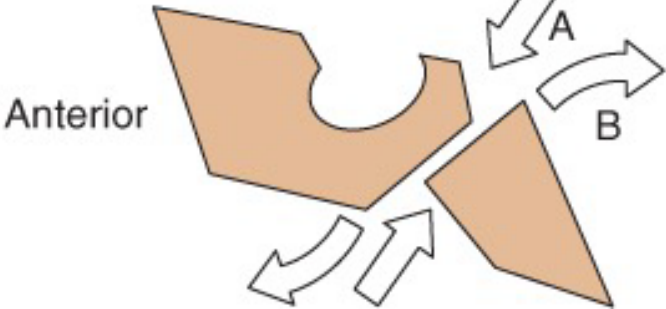


## Inferior vertical strain

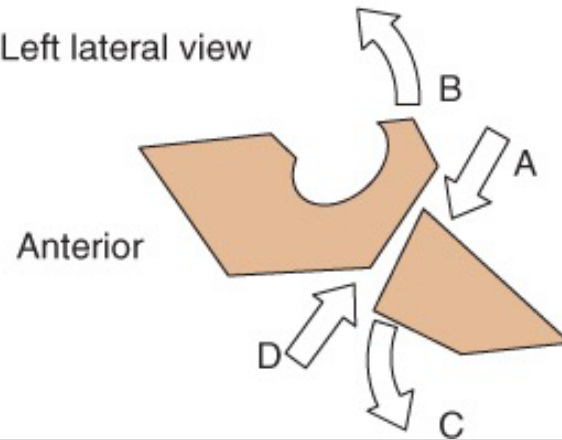
- Occiput rotates into relative flexion
- Sphenoid rotates into relative extension

# Putting it together - Vertical strains

Left lateral view



Left lateral view



Inferior vertical strain



Superior vertical strain



- Vertical Strains (Vertical Shears) – rotate same direction about their transverse axes
- Inferior Vertical Shear – both hands will rotate posteriorly, as if fingers are rotating up toward the ceiling
- Superior Vertical Shear- both hands will rotate anteriorly; or feels as if fingers are rotating down towards the table



# What strain pattern can be visualized here?

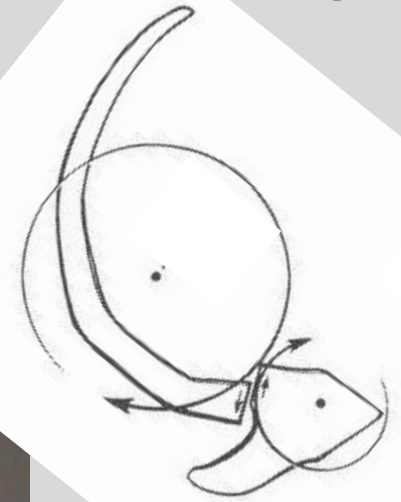


# Superior Vertical Strain



NEW YORK INSTITUTE  
OF TECHNOLOGY

of Osteopathic



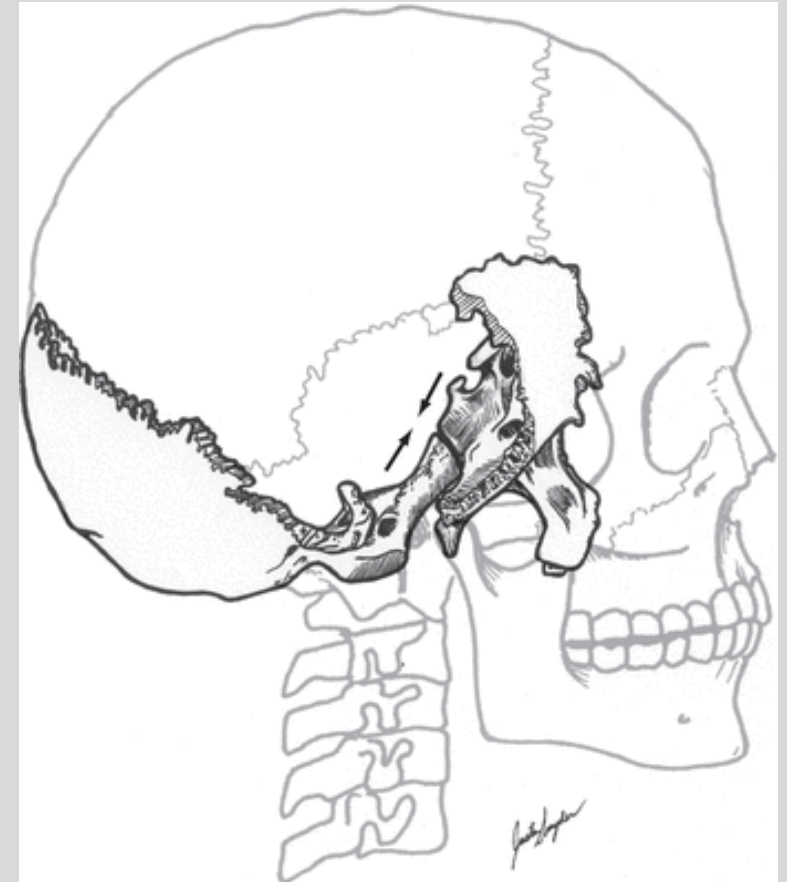
# Case #3 continued

- You palpate your patient's head and don't feel any motion present. Other than doubting yourself and thinking you are a horrible osteopath what else could be the reason why you don't feel anything?



# SBS compression

- Often secondary to trauma with a compressive force directed towards midline, jamming the SBS together.
- “Bowling Ball Head”
- Dense, compressed, relatively heavy when operator trying to lift
- Remember the most important aspect is the quality and motion of the PRM even if compression or other strain patterns are present

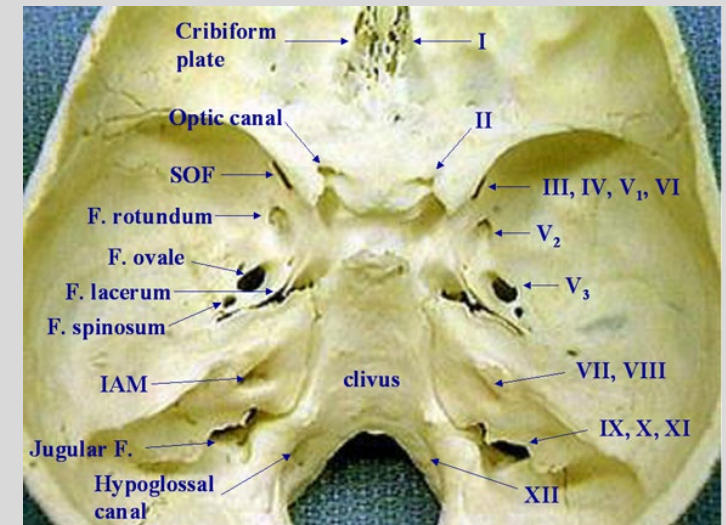
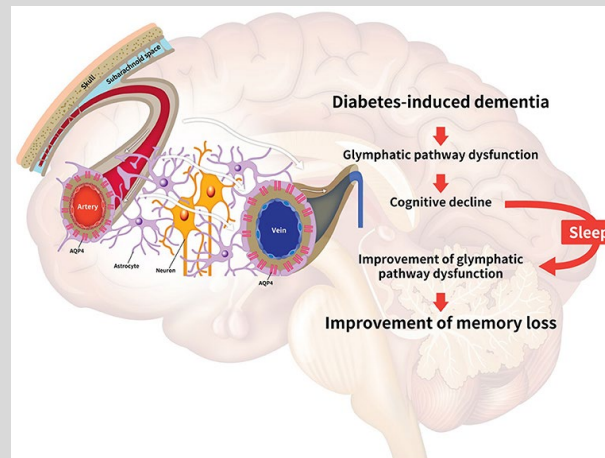
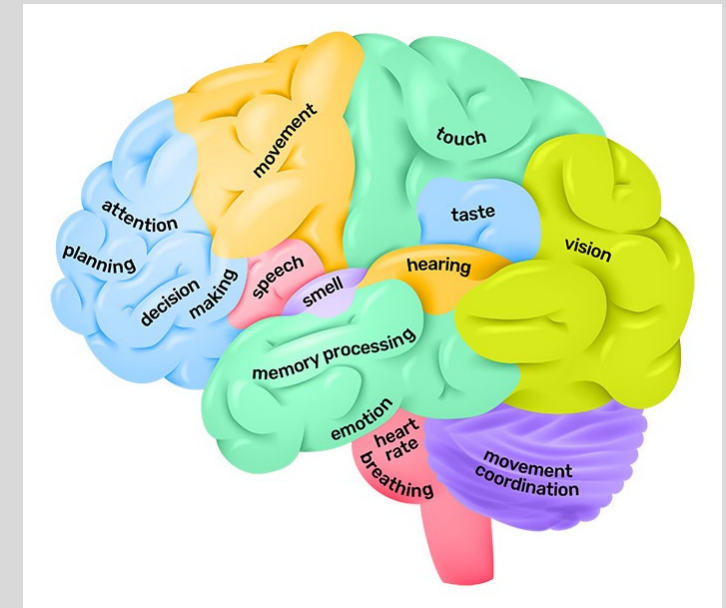






# Clinical Significance

- Diagnosis and treatment of cranial strains can help to restore natural motion.
- Reduce potential cranial nerve compression and balance ANS.
- Enhance circulation and blood flow.
- Improve glymphatic drainage.
- Improve sinus drainage.



# Contraindications

There are [absolute and relative contraindications](#) to Osteopathic Cranial Manipulative Medicine (OCMM)

- Absolute
  - Acute intracranial bleeding
  - Skull fracture
  - Acute cerebrovascular accident
- Relative
  - Coagulopathies
  - Space occupying lesion in cranium
  - Increased intracranial pressure



# Osteopathy in the Cranial Field (OCF) – Cranial Nerves

- Olfactory nerve – CN I (Cribiform plate – ethmoid)
- Optic nerve – CN II (optic canal – sphenoid)
- Oculomotor nerve – CN III (superior orbital fissure (SOF) – sphenoid)
- Trochlear nerve – CN IV (SOF – sphenoid)
- Trigeminal nerve – CN V (V1 – SOF, V2 – Foramen Rotundum, V3 – Foramen Ovale – sphenoid)
- Abducens – CN VI (SOF – sphenoid & petrosphenoid ligament)

# Osteopathy in the Cranial Field (OCF) – Cranial Nerves

- Facial nerve - CN VII (internal acoustic meatus – Temporal bone)
- Vestibulocochlear nerve – CN VIII (internal acoustic meatus – Temporal bone)
- Glossopharyngeal - CN IX (Jugular foramen - OM suture)
- Vagus – CN X (Jugular foramen - OM suture)
- Spinal Accessory – CN XI (Jugular foramen - OM suture)
- Hypoglossal – CN XII (Foramen magnum - OA & occipital intraosseous strain)

# What Problems Can Potentially Be Treated with Cranial Osteopathic Manipulative Medicine (COMM)?

## **Dental**

- \* Orthodontic Problems
- \* TMJ Syndrome
- \* Malocclusion

## **Digestive**

- \* Constipation
- \* Irritable Bowel Syndrome
- \* Nausea

## **Ear, Nose, and Throat**

- \* Chronic Ear Infections
- \* Chronic Tonsillitis
- \* Sinusitis

## **Genitourinary**

- \* Bedwetting
- \* Prostatitis
- \* Stress Incontinence

## **Neurological**

- \* Dizziness and Vertigo
- \* Head Trauma
- \* Migraine Headaches
- \* Post Concussion Syndrome
- \* Post-MVA Inertial Injury (Whiplash)
- \* Seizures
- \* Visual Disturbances

## **Pediatric**

- \* ADD, ADHD
- \* Autism
- \* Behavioral Problems
- \* Cerebral Palsy
- \* Colic
- \* Developmental Delays
- \* Failure to Thrive
- \* Learning Disorders
- \* Otitis Media (ear infections)
- \* Plagiocephaly (misshaped head)
- \* Spitting Up
- \* Strabismus (Crossed-Eyes or Lazy Eye)
- \* Sucking Difficulties

# What Problems Can Potentially Be Treated with Cranial Osteopathic Manipulative Medicine (COMM)?

## **Respiratory**

- \* Allergies
- \* Asthma
- \* Bronchitis
- \* Frequent Colds
- \* Recurrent Sore Throats

## **Somatic**

- \* Arthritis
- \* Fibromyalgia

## **Genetic**

- \* Congenital Malformations
- \* Down Syndrome

## **Orthopedic**

- \* Kyphosis
- \* Lordosis
- \* Postural Imbalances
- \* Scoliosis

## **Psychiatric**

- \* Anxiety
- \* Depression
- \* Post Traumatic Stress Disorder
- \* Sleep Disturbances

## **Pregnancy**

- \* Back Pain
- \* Groin Pain
- \* Varicose Veins
- \* Prevention of Labor Problems

Cited from The Cranial Academy Website <http://www.cranialacademy.org/benefit.html>

- Osteopathy in the cranial field (OCF) can be utilized for many different conditions.
- Understanding the motions and patterns that can occur at the SBS can help with diagnosing and treating your patients.
- Utilizing your understanding of the anatomy and cranial nerves paths will guide your treatment with COMM.
- Applying COMM can help support patient health and healing from injuries.

## Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>